

Fittr has helped transform the lives of people from all walks of life.
—**SUNIEL SHETTY**, actor, entrepreneur and fitness influencer

LOSE FAT, GET **FITTR**

THE SIMPLE SCIENCE OF STAYING HEALTHY FOR LIFE

MORE THAN
2,00,000
LIVES
TRANSFORMED

JITENDRA CHOUKSEY



LOSE FAT, GET FITTR

Jitendra Chouksey (better known as JC) is the winner of the BusinessWorld Young Entrepreneur Award 2020 and a first-generation fitness entrepreneur. He took a simple idea of helping his friends get fit over WhatsApp and turned it into a 100-million-dollar enterprise. Today, Fittr has trained and transformed more than 2,00,000 people from around the globe, provided livelihood to hundreds of people and, most importantly, has democratized fitness. With a community spanning over 104 countries and two million people, Fittr is one of the hottest Indian startups, backed by the likes of Sequoia Capital and Suniel Shetty. JC's vision for the community stemmed from his own childhood fears as an obese child who came from a humble background. He lives in Pune with his wife and daughter.

Follow the Fittr community on
Facebook ([@Fittrwithsquats](#)) and
Instagram ([@fittrwithsquats](#))

This book is for you if

you've tried to lose weight and failed every single time
you've managed to lose weight only to gain it all back
you've tried to gain weight but you're still thin
you've been teased and bullied about your weight
you've tried the General Motors (GM) diet
you've tried juice diets and detox cleanses
you've wasted money on supplements, pills and powders
your source for fitness information is the Internet
you used to feel good but now, you just feel (and look) tired
you've been making New Years' resolutions about fitness for as long as you can remember
you've been promising yourself to join a gym or go on a diet 'from next Monday'
you have a low sex drive and wish you could get your mojo back
you think you have 'bad genes'
you call yourself 'big-boned'
you hide behind large bags and long, flowy dresses
you wish you could fit into your college jeans again
you hate posing for photos because you're ashamed of how you look
you've bought at least ONE weight loss product on some home shopping network
you wear black because it's slimming
you look at a model's photograph and think, 'No one can look that hot without surgery'
you've seriously considered getting your stomach stapled
you hide behind Instagram filters because you hate the way you look
you look and feel older than you are
you think growing old and weak is just part of life
you know life will be better if only you got fit

you're tired of the way things have been
you're done making up excuses
you're sick of all the fake promises and fitness tips
you want to learn the method that has helped more than 2,00,000 people around the world
become healthy and stay fit
you're ready to change

Then...

THIS BOOK IS FOR YOU!

Praise for the Book

'If you've gotten your fitness advice from mainstream media without any results to show for it, JC's book will show you the way. JC does a good job introducing people to the wonders of calorie tracking and strength training in a simple manner. He debunks the most popular and damaging fitness myths and teaches you basic principles that actually work both in scientific research as well as practical application.'

Menno Henselmans
Founder of the Henselmans Personal
Training Certification Program

'JC does an elegant job of combining science, application, and stories to make this a book worth reading. If you're on a quest for the perfect body, then this book will teach you all the basics of exercise, nutrition, and supplements so you can start your own journey. Plus, JC busts a ton of common fitness myths so that you don't make the same mistake that so many other people have made. After reading this book, you'll feel like you're part of his family.'

Brandon Roberts
PhD, CSCS*D, TSAC-F

'In this much-needed book, JC not only busts the all-too-common myths that surround fitness, he also lays out a framework for the pursuit of fitness goals. By reading this book, you'll not only be able to guide yourself down the path of fitness, but you'll be able to do so in the safest, most efficient, and sustainable way possible. The best part is that you don't need even a bit of technical knowledge about biology or physiology, as JC speaks in plain language and explains complex concepts with remarkable simplicity. I highly recommend this book if you're thinking about getting fit and don't know where to start. If I had read it at the start of my fitness journey, I would have saved countless months of pointless struggle!'

Mike Israetel, PhD
Co-Founder, Renaissance Periodization

LOSE FAT, GET FITTR

THE SIMPLE SCIENCE OF STAYING HEALTHY FOR LIFE

JITENDRA CHOUKSEY



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FOREWORD

Mention the word ‘fitness’ and my ears perk up. It was in 2018, when I started hearing the buzz about a new fitness startup in India—Fittr. I took immediate notice.

India had plenty of gyms and dieticians that ‘guaranteed’ that you’ll lose weight fast. But this new buzz on the block was different.

The startup had taken a different approach to fitness. Instead of making tall claims and giving diet ‘tips’, they had taken it upon themselves to educate people about the science of nutrition and exercise, the twin cornerstones of good health. Their model was very simple: educate people about what works, teach them how to identify fitness myths and then, give them all the support they needed to make a lifestyle change and achieve their fitness goals. All advice in the community was given away for free, and the founder and his team went out of their way to respond to peoples’ doubts and questions.

Over time, the community attracted several thousands of people from all over the world—people who had tried to get fit several times in the past but hadn’t seen any lasting results. Once they found this community, they finally figured out what they’d been doing wrong and adopted a whole new fit lifestyle.

I was intrigued. And when I saw some of the transformations in the community, I was amazed.

As luck would have it, I received a call from the team asking if we could set up a meeting. I said yes, mostly because I wanted to meet the person who was the brains behind this community.

When Jitendra Chouksey walked in and shook my hand, I knew I had met someone who truly believed in what he was doing. He lived and breathed fitness but there was more to him. He was a man on a mission. When he looked me in the eye and said he wanted to make 50 million people fit, I believed him. I wanted to. I knew right away that this wasn’t someone who just says things, he walks the talk and has the courage of his convictions to stand by those words.

The book in your hands is a distillation of the knowledge that JC (as everyone calls him) has shared with his community over the years. With this knowledge, Fittr has helped transform the lives of more than 2,00,000 people from all walks of life. People who were lost, dazed and confused because they had fallen prey to random nutrition advice and followed fad diets who finally discovered, thanks to Fittr, that the science of becoming healthy is not at all complicated.

This is the book I wish I had found when I started my own fitness journey. My advice for you, the reader, is this: read and re-read this book until you know it like the back of your hand. If

your goal is to live a healthy life, you could not have asked for a better mentor and guide than JC and this book.

All the best!

Suniel Shetty
21 May 2021

PREFACE

Why should you read this book?

The short answer is that it can change your life. I don't blame you for being skeptical. You've probably heard that before, haven't you?

You've tried diets that were going to make you lose weight forever. You've bought gadgets that were going to give you a narrow waist. You've followed that fitness influencer who helped your favourite film star get to size zero and lower. You've done everything possible to get into shape but, so far, nothing has worked. Who knew it was so hard to get fit?

But I still stand by what I said. If you follow the method explained in this book, you **WILL** lose fat. You **WILL** build muscle and look hot. You **WILL** fit into your favourite clothes. And, most importantly, you will be healthy and fit for the rest of your life.

How can I be so sure?

I am sure because more than 2,00,000 people from all over the world have used the techniques explained in this book to become fit and healthy.

Many of them had been in the same position you are now—they had tried everything possible to get fit, but each time, they had failed. They had been on the verge of giving up, resigned to being unhealthy and unfit. But then, one day, they found a fitness formula that worked.

The same formula and techniques are explained in this book. They learned the principles, they followed the techniques and, most importantly, they trusted the process and took action. Life was never the same again. They were transformed.

What is possible?



Suman Saurabh



Before

110 kg



After

75 kg

Dr. Anirudh Deepak



Before

194 kg



After

83 kg

Sangeetha Thevar



Before



After

Shilpa Mehta



Before

85 kg



After

60 kg

Siddesh Vojhala



Before

140 kg

After



73 kg

Sonal Tyagi



Before

98 kg

After



58 kg

Anubha Pant



Before

96 kg



After

53.5 kg

One Day or Day One?

There are thousands of transformation stories that I can share with you, each one with its unique tale to tell, with challenges, difficulties, health issues, money problems. But the one thing they all had in common was the desire to change and become better. These are people who refused to play the victim card. They took action and found lasting health and happiness.

We've all made excuses at some point in our lives. When things don't go as planned, we find something or someone to blame. But ultimately, your fitness and health are in your own hands. You've made mistakes and fallen off the wagon before because you didn't know what works and what doesn't. In this book, my job is to show you what works and help you unlearn the myths and misinformation that have prevented you from getting fit.

It's not enough to merely have the right knowledge and tools. The only way to get fit is to use these methods with consistency. This book will teach you all that you need to know about getting fit and staying healthy but, ultimately, it is you who has to make the effort and put all that knowledge into practice.

Remember, fitness is the fruit of effort, not daydreams. Results WILL speak for themselves. So, bury the past. You've made mistakes, but that's alright. You're just human. Stop saying, 'One day, I will be fit.'

Today is Day One of your fitness journey. Welcome to the first day of a brand new chapter in your life.

INTRODUCTION

My Story

It was the year 2000 and Bollywood had just given us a brand-new superstar—Hrithik Roshan!

The guy was everything a small-town fat kid like me wanted to be: handsome, popular with the ladies, a great dancer and—most of all—shredded! To say that I was starstruck and obsessed would be an understatement. I even considered changing my name to Hrithik, but my horrified mother killed that dream in no time. So, I decided to do the next best thing—get a body just like Hrithik Roshan!

I got started, though I didn't have the foggiest idea about how I was going to build a body like that. So I went to my local gymwale bhaiyyas for advice. In hindsight, I wish I hadn't.

‘Drink 2 litres of raw cow's milk first thing in the morning.’

‘You're too fat! Stop eating rice.’

‘Goat meat makes you strong! Eat it every day.’ (Cue horrified mother once again!)

‘10,000 sit-ups and 15,000 push-ups every single day—that's how you look like Hrithik.’

I now know what terrible advice all of this was. But back then, I was just so grateful to receive any help. Budgetary constraints and dietary restrictions meant that milk and meat had to be replaced with dal and soya badi, but I followed everything else they told me to do. And to be honest, I did make some decent gains. I had age on my side and a lot of free time which I spent at the gym pumping iron. I made a decent amount of newbie gains, and I started to look and feel better.

Cut to 2013 and my childhood fantasies were behind me. I was now a grown-up and doing grown-up things: working in IT, living in another city and married. Life was settled. I tried to continue working out, but as my responsibilities increased, all that went out the window and I was back to square one! After all, between working a stressful job, balancing the household budget and spending time with the family, who can afford to build a good physique, right?

Moving to the big city also meant that I had to adopt its ‘modern’ lifestyle. I discovered pizza and burgers, and found that late-night films at the multiplex went well with soda and a large tub of caramel popcorn. Working odd hours in IT also meant endless trips to the coffee vending machine, and somehow, I'd always make it back to my seat with sandwiches and a bag of chips. Long story short, I packed on the kilos.

It was around that time that *Krish 3* was going to release, and everyone was talking about

Hrithik's lean-mean-ripped beach body look. He was giving everyone serious physique goals. I was bitten by the fitness bug again and decided to join a proper gym.

I didn't know it at the time, but my life was about to change forever.

Big City, Bigger Fitness Lies

So, there I was, the newest member of the swankiest gym in town. After all, a Bollywood bod needed a serious investment of money and time, no? I was ready to build those bulging biceps and abs of steel, and have everyone eating out of the palm of my hand in no time!

At my new gym, I was immediately accosted by half a dozen gym trainers who could tell that I was a new member, and they started their usual game: tell the newbie what terrible shape he is in but ‘worry not, main hoon na’. Of course, I had no intention of buying into their tricks, but I wouldn’t mind receiving some free advice and tips.

Turns out, fitness tips in the city are the same old myths, wrapped in fancier packaging.

‘Do fasted cardio in the morning, dude. The fat burn effect will blow your mind!’

‘Warm water, honey and a pinch of Sri Lankan cinnamon...’

‘If you’re not working out for at least two hours a day, just forget about getting that body, buddy.’

‘Heavy weights for gaining mass, light weights for losing!’

‘You should only eat egg whites. The yolks will make you fatter and give you cholesterol.’

‘Rice is like the WORST thing you can eat on a diet. Switch to brown rice or quinoa.’

‘Extra virgin olive oil, man! Ghee clogs your arteries and sticks to your fat cells.’

Once again, I didn’t question all this random-sounding advice. After all, those gym trainers looked like they had a lot of experience, and I was happy to receive such useful-sounding tips. I was sold every lie and I fell for each one of them, hook, line and sinker. I wanted to look sexy and shredded and I was willing to do anything.

I weighed 84 kg and had about 16 per cent body fat. I was told that if I wanted to look like you-know-who, I had to do two things: one, do cardio every single day, along with 2 hours of weight training, and second, I had to cut out all the bad foods and follow a low-calorie diet.

And guess what? It worked! In just three months, I lost 11 kg. Now, I know what you’re thinking. If this guy saw such great results, then why is he complaining about it?

Here’s the reason: while I went from 84 kg to 73 kg, what I lost was—you guessed it—muscle. But I didn’t just say goodbye to my muscles, I also lost strength, ruined my metabolism and ended up feeling terrible. My quest for a Bollywood physique had turned into a megaflop tearjerker. I had fallen prey to one of the most harmful things in the world: fitness myths.

What are fitness myths?

Go to Instagram and search for #fitnessinfluencer. Just pick any random profile and check their posts. Chances are, you will see any or all of the following:

- Booty pics in various stages of undress
- Potentially dangerous exercises such as bosu ball one-legged squats
- Plenty of talk about detoxes, cleanses and slimming teas
- Claims that so-and-so diet is the ONLY diet that works
- Entire food groups being demonized
- Claims that cutting out said food groups helped them lose weight
- Faithful groupies who praise everything that this influencer says and trash anyone who points out that they are spreading misinformation

Congratulations! You've just encountered fitness myths.

Science relies on research and uses testable explanations, but these fitness myths are a whole lot of half-truths and random tips that are continuously peddled without anything to back up the claims being made. Science asks questions and helps you gain clarity about matters that impact you; fitness myths want to mislead you so that you fall prey to some scheme or cure or must-have solution that leaves you more destitute in both health and wealth. Science is like the friend who wants to help you. Fitness myths are like the thugs who accost you in a dark alley, sucker-punch you and run away with your money and dignity.

These days, there are several forms of fitness myths and they can be found in all sorts of places: Dadi Amma's nuskey or tips and tricks, that wellness 'expert' and her TV show, social media handles or fitness columns, water-cooler gossip, your well-meaning but utterly clueless friend or relative, or as you've already seen, even your local gym trainers. Newspapers and mainstream media continue to be the biggest peddlers of such myths. After all, sensationalism sells. Who cares if people end up ruining their health, as long as someone makes money?

Well, I care. And I'm on a mission to do something about this epidemic of misinformation and outright lies.

My quest for truth

So, there I was, disheartened and still at 20 per cent body fat. What had gone wrong, I wondered? I had followed all the advice that my gym trainer had given me and yet I hadn't achieved my dream physique. Was something wrong with me?

I started looking for answers. I was like a man obsessed, and I had to find a solution.

This time, I did my own research. It's said that when the student is ready, the master appears. Out of the blue, I came across a research article which described exactly what had happened with me. I realized that literally EVERYTHING that I had been doing for the last three months had been wrong.

Too much cardio, very low calories and an 'all or nothing' mindset were clearly not the correct approaches. I was furious, both with my trainer and with myself. I had always prided myself on being someone who questioned everything and didn't take things at face value, but I realized that I'd been taken for a ride. My desire to see results had clouded my better judgment. But soon, anger gave way to curiosity. This incident became the wake-up call I needed. I decided to immerse myself in learning about fitness and nutrition. And that was the start of my journey.

Fast forward to 2021, and I am known as the founder of FITTR, one of India's fastest growing online fitness and nutrition companies. We are among the world's largest online nutrition and fitness communities, with more than 8,60,000 members. But the part that I'm the happiest and proudest of is the fact that FITTR has helped more than 2,00,000 people from across the globe get fit, lose weight and live happier lives. Altogether, they've lost more than 10,00,000 pounds of fat.

How did this happen? How did I go from fat to fit and even end up helping so many people? The answer is education. Education is the best weapon against the lies, distortions and myths that people like us have to face in our quest towards health and fitness. This book is a distillation of all my knowledge about health and fitness and I assure you that if you apply the principles and tenets that are covered in these pages, you will become fit and stay fit for life.

I wasn't an expert in health and fitness, but I was lucky to find teachers who showed me the right methods and helped me achieve my fitness goals. They've taught me an immense deal about what works and what doesn't, and that made all the difference.

Fitness is never about just losing weight: it's about gaining confidence, becoming healthy and disease-free and doing all the things that you've always wanted to do but couldn't because you were carrying around all that excess weight. As someone who used to be unhealthy and unfit, let me tell you that fitness will change your life. I promise you, you'll look good, feel great and start living your best life yet.

With the help of this book, I hope that you are able to discover the joy of good health.

HOW TO USE THIS BOOK

To get fit and healthy, the first thing you need to do is forget all the meaningless tips you've heard about nutrition and dieting. It's time to unlearn the myths and start with a plan that works. And, in this book, that's what you'll get to learn.

About This Book

This book contains everything you need to start your fitness journey and become healthy, presented in an easy to understand and step-by-step manner.

It is my sincere hope that you will read this book from cover to cover, practise whatever you've learned and share your knowledge with others.

It may be very tempting to skip the earlier chapters, but trust me, you'll miss a lot if you do so. Health is built step by step, brick by brick. It is a process and a journey. Don't be in a hurry to start running; first, learn how to walk correctly.

Don't get me wrong; I am not being condescending. But I have seen far too many people derail their fitness journeys because they didn't take the time to learn the basics. You may find the going easy in the beginning, but as your journey progresses, you will need guidance and assistance, which may prove difficult if you've skipped the vital initial steps.

What You Will Learn

Getting Started: This chapter teaches you the first principles of fitness and lays the groundwork for the rest of the book.

Fitness Is Science: In this chapter, you will be introduced to metabolism, energy balance, quantified nutrition and exercise.

Common Fitness Myths: This chapter covers some of the most outrageous fitness myths that have been floating around for ages. You might even believe a few of them. It's time for a reality check!

Nutrition: It is time to learn the real meaning of nutrition and healthy eating. This chapter delves deep into macronutrients, micronutrients and hormones.

Prepare Your Own Nutrition Plan: After learning about diets and nutrition, it's time to prepare your personal nutrition plan. You'll also get to understand some popular diets and see how to use them.

Exercise and Training: This chapter deals with the basics of exercise science and an introduction to different exercises, while helping you find the right type of exercise for your fitness goals.

Prepare Your Own Workout Plan: In this chapter, you will put the principles of the previous chapter into practice, designing your personal workout plan to ensure your progress. And you will also get the answer to the big question—should you do cardio?

Supplements: You will find out which supplements work and which ones to avoid. This chapter includes a special section on Ayurvedic supplements.

Parting Words: Now that you've read the book, where do you go from here?

I'm excited to share my knowledge with you through this book. Let's learn some fitness science, let's get educated and let's kick some myth-ass!

GETTING STARTED

Some time back, I attended a unique Facebook live event: my college friend was hosting a ‘Watch My Son Take His First Steps’ watch party and everyone was invited.

Let me tell you, the show was full! It seemed as if everyone he knew—parents, uncles and aunts, friends, colleagues, friends of friends—wanted to share in this experience. The child was blissfully unaware that he was the centre of attention. He took his own sweet time getting into the act. He lolled and rolled around even as his excited parents cooed and coaxed and encouraged him to stand up. Eventually, he decided to treat the audience to the main show. He rolled over, crawled to the centre table and with a grunt and a giggle, got onto his knees. Supported by Papa, he slowly raised himself, and after a few hesitant moments, he gurgled for the camera and put one foot in front of the other.

The comments section erupted with hearts, and everyone chimed in with their congratulations for the parents. Their son had achieved a significant milestone.

Later that day, I sat and thought about what I had just witnessed and related it to my own story. From being a fat kid living in a small town, to getting in the best physical and mental state I have ever been, and then creating my own company with the experiences of my fitness journey.

It has been a long journey, filled with failures and doubts but also dreams and accomplishments and loads of excitement. But it had all started with that first step. And isn’t that where all of us have to begin?

Fitness is like running a business or getting rich. You first have to understand the basics. Then, once you have the knowledge and skills, you take the first step and start putting all that theory into practice.

But do you become a success overnight? No. You have to put in the hours and slog for weeks, months, sometimes years before you achieve your goal. All those ‘overnight successes’ are anything but that. Trust me, I know!

You need to build a system that works; you need to understand the process and the various steps involved. You will eventually make progress, but you can’t stop there and be satisfied with what you’ve done. The only key to success is to keep going, get better, and improve every single day. Whether you’re building a world-class organization or creating a healthy and fit life, it is a continuous process, and you have to turn up every single day and do your bit. A millionaire needs to work on staying a millionaire. If they stop, then they lose their wealth and slide backwards.

Fitness, too, is a two-way street. Just because you've gotten fit doesn't mean you'll always stay fit. It's quite easy to fall right back into your old habits and behaviours and lose your gains and health. But if you understand the basics, keep applying what you've learned and keep striving for incremental improvements every single day, you will not slide back down that hill, ever.

But, it all starts with that first step. And this book is your first step. This chapter and the next one are all about the basics of fitness. These first steps will help you understand the fundamentals of fitness. We'll be learning some must-know principles, covering some key concepts and laying the groundwork for the new YOU—fitter, better and definitely smarter.

Some of you may be tempted to jump straight to the later chapters. I would urge you to stay on and read this chapter till the very end. In fact, re-read it and keep reminding yourself of the principles that you'll read here.

Far too many people have come to me and said, 'I wish I had known all this before. I would have become fit a long time ago!' This book is going to equip you with all that you need to know to become fit and stay fit for life, and it all begins with the principles you will learn in this chapter, so don't skip it.

Let's begin!

First Principles

- **Health and Fitness:** A healthy person is someone who is integrated at all levels—physical, mental, emotional and social. Fit is simply an expression which is used to evaluate different parameters related to your health and well-being. Physically fit, mentally fit, financially fit, socially fit: all these health parameters are tightly integrated. You can't be physically fit if you're not mentally fit and you can't be mentally fit if you have physical ailments. To think that you can excel in one of the health parameters while ignoring the rest is an illusion. All of them are equally important.

This book will cover all aspects of physical fitness. And trust me, only a fit body can house a healthy mind, fully express healthy emotions and contribute to your overall well-being.

- **Fat Loss Over Weight Loss:** When you step onto the weighing scale, the number that stares back at you is a combination of your fat mass, muscle mass, water, skeletal system and organs. If that number alarms you and you decide to do something about it, remember that aiming for weight loss isn't the best way forward.

Be clear about what you want to accomplish on your fitness journey. Aim for fat loss.

- **Find a Lifestyle Fit:** People make this mistake all the time: they find out about some new diet or workout program and decide that it is *the* thing that will make them fit and fabulous. So, if their favourite fitness influencer adds coconut oil to their black coffee or works out twice a day to make 'booty gains', then so must they. Never mind the fact that they hate the taste of coconut oil and haven't seen the inside of a gym for ages.

Your fitness choices—diet, exercise, rest—must fit into your current lifestyle, not the other way round. I'm all for adopting a healthier lifestyle but it is essential to realize that change is a gradual process, and your chances of success increase when you set the right expectations and work towards them. Trying to fit someone else's schedule into your life may work for some time, but the chances you'll fall off the wagon are high. Find what works for you and do it.

- **Adherence Is Key:** This principle is related to the previous one. Rome wasn't built in a day. You too shouldn't expect to undo the damage of years of poor food and exercise choices in just a matter of days or weeks.

Any good transformation requires two things—visualization and realization. Think of yourself as an architect who's planning a building. Can you build something without laying out a plan or having a picture in your mind of the finished building? Similarly, first, you have to visualize what you want to look like and who you want to be in your life. Then comes the realization. If the plan is ready, it's all about doing the same thing over and over again, laying bricks over and over again till you complete the foundation and then a floor and then several floors.

Getting fit requires time, hard work and motivation; simply changing your diet for a week or becoming a weekend warrior at the gym won't help. Diets take time to show results and workouts take time before you become beach ready.

Transformations don't happen overnight: just ask any of the 2,00,000 plus people who have been transformed by Fittr. They had to put in the hours and make consistent efforts. Just like them, you too will have to ignore the voice that tells you to quit. You can't let boredom and lethargy set in. You will need to say no to parties and all-you-can-eat midnight buffets and pained requests from your favourite aunt to 'just eat one laddoo—mere liye' and that's why it's crucial to have a visual of your future self in your head.

Want to get fit? Eat right. Work out. Rinse and repeat. Staying on the course is the only thing that works. You won't regret it.

- . **Fitness Is Science:** If you've been getting your health tips from the lifestyle section of a newspaper or health show or your friends and relatives, then the chances are that what I've written in this book will be brand new information to you.

There is a severe shortage of scientific information in popular media and sadly, that's the first place people turn to when they decide to get fit. Social media isn't any better. Instagram, Facebook and YouTube are flooded with fitness influencers who sell everything from fad diets to magical supplements. Even those who have workout videos seem to be more interested in bending over for the camera than in wrapping their heads around any actual fitness science. The ones who have good content and who talk science are in the minority. Apparently, science isn't sexy enough.

Well, my sincere hope is that after you've finished reading the book and applied its learnings in your own life, you will do your bit to spread the word of science and help someone else take the first step towards a fitter life. In the next chapter, we'll look at some of these scientific principles of fitness that have stood the test of time and rigorous research.

FITNESS IS SCIENCE

Correct me if I'm wrong, but don't you think that people take better care of their cars than their own bodies? Not only that, they even seem to have a better understanding of how a car works than the inner mechanism of the bodies they inhabit.

What most people don't seem to understand, however, is that if your car breaks down or stops working, you can always get it repaired or replace it with a new one. But you can't do that with your body: this body is the only one you have and ever will have, you can't cast it aside and upgrade to a newer model. And yet, people treat their bodies as if they're dispensable and easily replaceable. Remember: if you damage your body, you'll have no choice but to live with it.

I use the car analogy to explain how the body works, in the hope that, by seeing the similarities between a car and the human body, people will start treating their body with the care and respect it deserves.

Think of your body as a car. What makes your car run? Fuel. You can't drive your car if your fuel tank is empty, can you? Just as you need to fill up your car's tank with fuel, you also need to provide your body with the right fuel so that it can keep working. This fuel is the food that we consume, the nutrition that we offer ourselves.

Now, if you have fuel in your car but the battery is dead, will the car run? No. In the same manner, merely eating food won't do anything for your body unless there is a mechanism to utilize that food and convert it into energy. That mechanism is your metabolism.

Metabolism

Every living organism has an in-built laboratory where several chemical reactions and processes take place all the time. Broadly, these processes have three functions:

1. Conversion of food into energy
2. Conversion of food into the building blocks of the body—cells
3. Elimination of waste from the body

The process of converting food into energy is called catabolism. It involves breaking down the food molecules into smaller units so that they can be oxidized and converted into energy. Some of these smaller units are then recomposed into cells within the body. This building process is called anabolism. When muscles repair and grow, anabolism is at play. Metabolism is also responsible for digestion and for transporting substances to the cells of the body.

Naturally, these processes generate some waste which is then eliminated from the body.

Every organism has a metabolism that keeps it alive and enables its organs to function and grow. Like a car, the human body has exceptionally efficient systems for utilizing energy, something that even German engineering technology could not have created. At different points of time and depending on the level of activity you are performing, the body needs a differential amount of energy to keep it running smoothly.

Let's say, you're home on a Saturday. Your spouse and children have gone out, so you're all alone at home. You decide it's a good time to catch up on some Netflix. So, you lie down on the sofa, put your feet up and before long, you've dozed off.

Suddenly, you get a call from your spouse—their parents are coming over for dinner. You need to hop over to the supermarket and get some supplies. Groaning to yourself, you stand up, put on your shoes and rush out of the door. The supermarket is just around the corner, so you decide to walk there instead of driving.

In each case—lying still, standing up, walking—your body is using energy, but the rate at which energy is being utilized differs every time. When you're lying down and doing absolutely nothing, you're still doing a lot of things. Confused? Even though you don't know it, your body is ticking away like clockwork. Your lungs are drawing in air, your heart is pumping blood, and your brain is still functioning. All these things keep you alive and the energy needed to run all these essential functions, even when your body is in a state of rest, is known as **Basal Metabolic Rate (BMR)**.

It's evident that when you start walking, you will use energy at a rate higher than your BMR—the more intense the activity, the greater the amount of energy being utilized to fuel that activity. When you consider all the physical activities you do throughout the day and add up the differential energy needed for them, you arrive at your **Total Daily Energy Expenditure (TDEE)**.

There is a third concept that you need to know about, the **Thermic Effect of Food (TEF)**. When you consume food, your body needs to utilize a certain amount of energy for digesting or metabolizing that food. This is TEF.

If we were to take a rough average of how our bodies utilize energy, BMR is 70 per cent of all energy used, followed by physical activities, which use 20–25 per cent of energy, followed by TEF at about 10 per cent. The sum of all these is TDEE.

Calories

For cars, fuel provides the energy to make them run. But can you fill diesel in a petrol car and expect it to work? Of course not! Cars, in general, are built to derive energy from only one type of fuel (unless they're hybrid cars).

The human body, on the other hand, has multiple sources of energy. This energy is derived from the food you eat. Different foods provide different amounts of energy to your body, which is measured in 'calories'. We will go into the details of the food groups in the next chapter, but for now, this is what you need to know:

1 gram of carbohydrates=4 calories

1 gram of fat=9 calories

1 gram of protein=4 calories

Metabolic Regulation and Energy Balance

Now, we're getting to the main part. Hold on to your hats!

There's a lot happening inside your body—chemical processes, essential functions that keep you alive, and activities that need energy, which must be generated as and when required.

We live in a time when we don't have to worry about our next meal, but that wasn't always the case. Our hunter-gatherer ancestors lived a nomadic existence. Food was scarce and not always available when they needed it. So, our bodies had to develop systems which were energy efficient.

The human body likes to maintain a balance between the energy it receives and the energy it uses (termed energy homeostasis). In other words, in an ideal situation:

Energy received from food=Energy utilized for all activities

or

Calories in=Calories out

But, this doesn't always happen.

Why Do We Get Fat?

Your parents-in-law come home for dinner and bring some sweets or mithai. You sit down to a hearty meal and have some of the mithai as well. You usually only have one piece, but tonight, your mother-in-law insists that you should have more. You can't (or don't want to) say no, so you gulp down three pieces of the melt-in-your-mouth sweets.

It's a Saturday, you've done nothing but sleep all day, and now, you're wolfing down sweets. Today, your energy balance is off. That's because:

Energy received from food > Energy utilized for all activities

or

Calories in > Calories out

Your spouse, on the other hand, has had a very hectic day. Shopping, cooking, helping the children with homework, they have been running around all day. And, unlike you, they don't have a sweet tooth, so they stick to ghar ka khaana or home-cooked food.

Now, in your spouse's case as well, the energy balance is off but in a different way. For them:

Energy received from food < Energy utilized for all activities

or

Calories in < Calories out

Your body may be like a car, but unlike a car's fuel tank, you don't have a limited capacity. Try filling 25 litres of fuel in a 20-litre fuel tank. You can't; the excess will spill out. Sadly, your body has no choice but to do something with the excess calories that it has received. So, it stores the calories, to be used another day to provide energy for your activities.

But, in your case, that day never arrives. Day after day, week after week, you keep on eating more calories than you burn. All those excess calories keep piling up, and your body goes on storing them. When your parents-in-law see you after three months, your face is looking rounder than it was the last time. Your clothes have become tighter and when you sit down at the dinner table, you have to loosen your pants.

You, my friend, have gained weight. And this has happened because you've been consuming too many calories. If you want to lose weight (actually, you should want to lose fat), you will either have to reduce the number of calories you consume or increase your level of activity so that you burn more calories than you consume.

In other words, if you consume fewer calories or expend more calories than your TDEE for a sustained period, you will lose fat.

That Is All

Simple and logical. And yet, we have managed to complicate matters so much. We find ourselves unable to apply this simple principle and then blame our weight gain on everything—from our sluggish metabolism to the foods we eat.

How many of us have been told that carbs make you fat and thought, *‘I should stop eating rice, then I’ll lose some weight’*?

How many of us have tried to lose weight by drinking a glass of warm water with lemon first thing in the morning? Apparently, that is supposed to melt belly fat.

Your friend told you to start eating avocados because they contain healthy fats and, after all, fat burns fat. You ate avocados with breakfast, lunch and dinner and guess what? You gained 2 kilos in a month!

Did it ever occur to you that maybe, just maybe, you should eat less and move more? Of course, it didn’t. After all, how could such a simple explanation be the right one?

What's the Solution?

So, you've finally seen the light. From now onwards, you will watch what you eat and also exercise.

But 'eat less' and 'get some exercise' are random pieces of advice. To make them work, so that they have a lasting impact on your health, you need to follow a scientific approach to diet and exercise.

Quantified Nutrition

Quantified nutrition means a diet protocol where you measure all the food that you consume and keep track of your calories. You will no longer eyeball your food and guesstimate how much you're eating. You will actually *measure* your food intake.

I can sense you staring at those words in disbelief. 'What? Measure all my food? You mean use a weighing scale like a grocer and check the weight of all my food?'

Well, we measure almost everything from money in the bank to our weight on the scale. But somehow measuring food makes people scratch their heads in disbelief as if it's something odd. You think chefs don't do it? Of course, they do, it's just that they don't do it to stay fit, maybe because they don't know what you're about to learn.

I hear you. You've never measured your food before, and you're worried that it's just too difficult. But trust me, it's not as difficult as it may sound. I'll show you exactly how you can easily make this a part of your daily routine. Besides, this isn't a new idea at all. We measure our food all the time. When we add masalas and salt to our food while cooking, when we add 'just a drizzle of olive oil' to our salads, even those times when we say we're going to eat 'just a tiny piece' of cake or one katori or bowl of chawal, we are measuring our food. But here's the problem: 'drizzle', 'tiny piece' and 'katori' are quantities which contain calories. And unless you have precisely measured how much food you are shoving down your throat, you won't know how many calories you are consuming.

The science behind fat loss is very clear: eat more calories than you burn, and you *will* gain weight. But don't worry. By the time you finish reading this book, measuring food will become second nature for you.

Case Study

One of my dearest friends is Ali Engineer, a restaurateur who makes the most delectable food I've ever eaten. Naturally, surrounded by food all the time, he piled on the kilos like nobody's business. While his wife would pack him a nice, healthy meal every day, Ali would invariably end up eating food at the restaurant. 'A chef's job is to make sure people leave his restaurant with a smile on their face and good food in their belly. I *have* to taste all the food; otherwise, how will I know?'

Well, what do you know? By his thirty-eighth birthday, his weight had ballooned to 108 kg. He was forced to wear 46-inch trousers and XXXL tents as shirts. And on top of that, he was borderline diabetic, had high cholesterol and was always exhausted. His worried wife took him to several 'dieticians' who prescribed a wide variety of solutions: there was the 'Eat Every 2 Hours to Boost Your Metabolism' diet, the '2 Digestive Biscuits with Half a Cup of Tea' diet. One dietician even told Ali to follow the GM diet. Naturally, nothing worked.

I met Ali at his lowest physical and emotional point, and the advice I gave him is precisely what I am telling you now—start measuring your food. That is the only way you can know how many calories you are actually consuming. Using some basic principles, Ali lost a whopping 44 kg and is down to 64 kg, the lowest weight of his entire adult life! And it was all thanks to quantified nutrition and resistance training.

Ali Engineer



Before

108 kg

After



64 kg

Exercise

Food is one part of the fat-loss equation, the other being exercise.

Remember BMR and TDEE? We had seen how physical activity is fuelled by about 20–25 per cent of the available energy.

Let's take the case of two people, Vijay and Rahul. Both of them have a BMR of 1,500 calories, a TDEE of 1,800 calories, and both of them consume 2,000 calories.

Vijay is a certified couch potato. He simply will not move a muscle. He has a very sedentary lifestyle, and even when he comes home, he just plops himself in front of the TV, watches the news, eats his dinner and goes to bed. Rahul also has a sedentary lifestyle—most of us who have desk jobs fall into this category. However, he has just started lifting weights in the gym, at least 3–4 times a week for 45 minutes to an hour each time.

It should, therefore, not surprise you if I told you that six months later, Vijay has become fatter. His TDEE was 1,800 calories, whereas he was consuming 2,000 calories. Every day, he consumed 200 calories more than he burned which means in six months, he consumed 36,000 excess calories. But what about Rahul? Since he was working out, his TDEE increased from 1,800 calories to 2,200 calories. He was consuming 2,000 calories per day, which means a deficit of 200 calories per day or 36,000 calories in six months. Naturally, Rahul lost fat.

But there's something else that also happened. Vijay did not work out, and so, he gained fat. Rahul consumed calories below his TDEE and worked out, so he lost fat, managed to preserve his muscle mass and gained strength. Vijay looks fat and unhealthy, whereas Rahul looks lean and more muscular.

I hope this example has driven home the point I'm trying to make: if you want to get fit and stay fit, then you need to watch how much you eat and also incorporate physical activity into your daily routine. But not just any physical activity—one that gets you moving, builds strength and doesn't result in muscle loss.

Do I Really Need to Work Out?

A lot of people who want to lose weight (there's that word again!) decide that all they need to do is simply start eating less food. They wake up one fine day and decide that they'll stop eating rice (because 'rice is evil'), have only one chapati for lunch and another for dinner, eat lots of salads and wash it all down with some healthy fruit juice or watery, fat-free buttermilk (because 'fat is evil too!'). Maybe they'll also go for a walk in the morning or do some yoga and Bhastrika Pranayama because it's 'humari parampara' or 'our traditions'.

And hey presto—they've lost weight!

Now, the bad news: this will ruin your health. The longer you subject your body to this 'diet', the more metabolic slowdown you'll incur. Ideally, you should always eat above your BMR. If you don't, you will deprive your body of the energy it requires for running even your most fundamental processes. Tiredness and dark circles will be the least of your concerns—you'll end up nutrient deficient and sick in the long run.

You may think I'm being melodramatic. I wish that were true. I've met so many people who have ruined their health because they've been consuming very low calories, which leads to hair loss, low sex drive and they bounce back to their previous weight or worse.

You can only lose so much weight by cutting calories. A much better strategy is to increase your activity levels and find your caloric sweet spot, which should be more than your BMR but less than your TDEE. This is your fat burning zone.

So what type of exercise works? The answer is simple: lifting weights, aka resistance training. If you want to burn fat, build muscle and look good (with or without clothes), then lifting weights is *the* best thing to do!

But what if you don't have access to a gym? Don't worry; we'll be looking at some great options for you to break a sweat, burn some calories and get super fit, all in the comfort of your home.

Should Women Lift Weights?

Do women have muscles? They do? So, what's the question again?

One of the oldest myths circulating in fitness circles is that women should not lift heavy weights. And if they do, they'll turn into a she-Hulk or end up looking like some Soviet-era Steroid Sally.

Gym trainers and fitness influencers will tell you that women should aim to 'look toned'. Usually, that means a combination of eating an ultra-low-calorie diet and lifting lighter-than-light pink dumbbells. Women and girls have been told that they are the weaker sex, that lifting will make them manly and—horror of horrors—who will marry them then?

Our grandmothers used to lift weights every day. The grinding stone in my ancestral home in the village weighed a ton, and every morning, my 'weak' grandmother would lift it in one clean sweep, carry it to the veranda and mix the day's spices. Every week, she would pour red chillies into a stone mortar and pound them into a fine powder with the help of a heavy metal pestle. She did this as a young bride, as a mother of five and continued with this routine well into her 70s. And let me tell you—she was fit, supple and not the least bit manly! If she can do it, then so can you.

Case Study

Karuna Khatwani is a 57-year-old grandmother who had been told the same thing—women should not lift. A wife and successful businesswoman, her health had been steadily declining for years. After suffering severe back injuries, her health deteriorated even more, and she weighed 77 kg. Desperate to try anything to lose weight, she stepped out of her comfort zone and decided to start lifting weights. Today, she's lost 14 kg and is one of the most dedicated people in her gym. Come hail or foul weather, Karuna makes sure she doesn't miss a single workout. And as you can see, she looks nothing like the Hulk!

Karuna Khatwani



Before

77 kg



After

63 kg

Now that you've chosen to stay on this ride till the very end, I promise that your life, health, body, energy and fitness will be vastly improved and you'll become a whole new person.

COMMON FITNESS MYTHS

I couldn't believe my ears.

'Yoga changes the molecular structure of your body. If you believe you can't get fat, you won't!'

Thunderous applause greeted this astonishing statement made by a fitness influencer-turned-new-mom. The crowd wanted to know how this megastar went back to size zero post-pregnancy, and this was the nugget of wisdom that she delivered.

Sadly, she isn't the only celebrity making such tall claims. It turns out that these days, just about every famous and semi-famous person has an opinion on everything from the economy to India-Pakistan relations to fitness. Well, they're free to say what they want about other subjects, but fitness is a whole different ball game. Here, what matters is science.

Do You Even Science, Buddy?

Sadly, not everyone understands science, nor do they care. People make snap judgements about other people based on how they look, and if you look young and hot and are dripping diamonds from head to toe, then people will tend to trust you and look up to you. Science is a lot less sexy (at least to them). I mean, why would you want to make the effort to find out if such statements are backed up by research or not?

There's a reason why I've kept this chapter towards the beginning of the book: I want you to know what you're up against. The lies, the fabrications and the outright misleading statements designed to make you part with your money (and your sanity) as you go on a quest to get fit.

You're one of the lucky ones—you've found this book! Think of the countless unlucky souls who were fooled over and over again, and still aren't fit. Read these lies and remember them. Then, once you've finished reading this book, and finally know the science behind fitness and health, I want you to come back and re-read this chapter. I want you to shake your head in disbelief. I want you to roll your eyes, throw your head back and laugh.

And the next time someone mouths these 'facts', I want you to explain to them why they're wrong. Here are just 10 of several myths that this book seeks to demolish. Let's get started!

Myth #1: Warm water with honey and lemon helps you lose weight

This is one of the oldies that is still making the rounds of fitness websites, newspaper columns and health blogs everywhere. I remember one article which claimed that the hotter the water, the faster your fat will burn. I know what's going to burn when you drink a cup of boiling water and it ain't your fat cells! And if a good burn is what you're looking for, why not try acid?

Here's the truth: water keeps you hydrated. Almost 70–80 per cent of your body is water. Without water, a number of your body's vital functions (and organs) would not work. In short, water is life. But it is not magic.

Warm water with honey and lemon sounds like something you might drink if you had a cold. And, depending on the size of your bladder and the weather conditions, your body will eliminate it either as sweat or urine. And, let me assure you, no fat cells will be exiting your body with the water.

Myth #2: Eating rice at night makes you fat

Another myth that's been around for at least as long as I have walked the planet. This myth is tied not only to the next one about carbohydrates, but also about white foods in general. All white foods are blamed for causing obesity. The reason? Sugar is white; therefore, it is processed, and hence, causes obesity. Rice is guilty by association since it is both white and processed.

When I was young, my mother used to tell me, 'Jeetu, don't lie down right after eating rice. You'll become fat.' I think the myth came about because rice makes you feel full for longer, and if you overeat, it can cause bloating.

As you'll see later, it's not rice that makes you fat (no matter when you eat it).

Myth #3: Protein damages kidneys

Protein is considered a building block of the body. Then, how did it get accused of causing kidney damage? Protein is composed of nitrogen, and one of the by-products of protein digestion is urea. The more protein you eat, the more nitrogen is excreted, and therefore, the more urea is generated.

Usually, in an average body, this urea gets flushed out of the body through urine and excreta. However, in some cases, the urea does not get flushed out completely, which could happen due to the malabsorption of protein or some other issues. As a result, this urea turns into uric acid and raises the pH levels of your body. If these conditions persist over a prolonged period, then they could lead to gout or the formation of kidney stones. However, this is NOT on account of protein.

As you will see in later chapters, protein is a critical macronutrient, and there are clear benefits to consuming protein.

Myth #4: Carbs make you fat/fat makes you fat/any food makes you fat

I've clubbed all these myths together because they're all the same type of lie—the demonization lie—where one particular food item or one category of food is blamed and held responsible for making us fat.

Isn't this exactly how humans behave? We even demonize religions and entire communities and hold them responsible for all the evil in the world. After all, this is just food. People who make these kinds of claims aren't trying to help you; they're out to make a quick buck at your expense. There is an ulterior motive to peddling such lies—to make you buy their product which, naturally, does not contain the guilty food item.

So, you have fatless butter (because fat makes you fat) and wheatless rotis (made from super expensive almond flour because carbohydrates make you fat). Before you know it, you've started preparing all sorts of strange recipes where entire foods or food groups are eliminated. The result isn't a fitter you. In most cases, it is a poorer you because you've been convinced to embrace an entirely bogus lifestyle.

Myth #5: So-and-so food makes you fit, healthy and sexy

This myth is related to Myth #4. Just as no single food or food group can make you unhealthy, no single food or food group can make you healthy, at the exclusion of all others. The key to good health is quantified nutrition, and it does not discriminate against any foods.

Now that we've come to the halfway mark, let's switch gears and talk about some homegrown, Indian myths, shall we?

Myth #6: Festival food doesn't make you fat

India is the land of festivals. I have nothing against festivals: they bring people together and spread joy. My only problem is that people stuff their faces with food under the pretext of enjoying a festival. Festivals have become an excuse for gluttony. People overeat for days and then wonder why their clothes won't fit.

Enter Enlightened Being aka Celebrity Fitness Expert who made this revelation to the masses and had them bowing to her and had me pulling out my hair in disbelief: 'Shrikhand puri eaten on festivals won't make you fat!'

If this didn't make your brain freeze, here's her strategy: 'Prepare your mind for the fact that you will be eating festival food. If you tell your mind that it needs to be ready for the inevitable hog fest, then it will make peace with that fact and won't get stressed out. If you eat with a calm state of mind, your food will not get converted into fat!'

I will dip my Under Armour cap in ghee and eat it if she or anyone else can prove to me that this is possible. The worst part is that people were defending this hare-brained notion. The need for science—and this book—became more apparent to me than ever before!

Myth #7: So-and-so works because of Ayurveda

Ayurveda is an ancient science. It contains a detailed and deep understanding of the human body and mind that is awe-inspiring for its sheer depth and scope.

The same cannot be said of people who will attribute anything to Ayurveda and say, ‘Shastron mein likha hai! (It’s written in the scriptures!)’ Therefore, it is the truth. Never mind that they have neither read nor heard of *Charak Samhita* and *Susruta Samhita*. There are very few things I hate more than someone claiming to know something about Ayurveda or science and then not backing it up with facts.

Also, no disrespect to anyone’s grandmother but Dadi ke nuskey or grandma’s tips should probably have gone to the grave with her. This book is about facts. You’ve been warned!

Myth #8: Momos in Manipur won't make you fat but momos in Mumbai will

This is also known as 'Think Global, Eat Local', another unscientific new-age fitness homily.

Apparently, a momo is a fat burner when you eat it in Manipur, but the same momo becomes a fat bomb in Mumbai and makes you gain weight. The reason? Our bodies lack the enzymes that can effectively break down a momo and so, it will immediately make a beeline for your fat cells.

Let's say you make chicken momos using 20 grams of chicken and 20 grams of flour. Maybe you use 5 grams of oil to seal the dough. Then, you steam them and eat them. In Manipur, the momos will have 205 calories. Now, if you were to prepare the same momos in Mumbai using the same ingredients and cooking technique, how many calories would they contain? Still 205 calories.

I hope I have made my point. If you still don't get it, then you definitely will after you read the section on 'Energy Balance'.

Myth #9: Always eat what your forefathers ate

Charles Darwin just rolled over in his grave. The evolutionary history of humans goes back many millennia. And yet, if a Punjabi were to eat idli sambhar and a Tamilian were to eat aloo paratha, they would supposedly face digestive issues which would lead to—you've guessed it—nutritional deficiencies and fat gain!

Idli sambhar and aloo paratha contain carbohydrates, fats and a little protein. Some of the constituent ingredients—dal, potatoes, rice, wheat and ghee—are also very similar and in fact, often eaten as substitutes for each other. And yet we are supposed to buy into this lie that only those foods that our forefathers ate will provide us with the nutrition we require.

Myth #10: Women should not lift weights

I've already written about my old grandmother who lifted heavy weights all her life, albeit not barbells and dumbbells. This argument, that women who lift weights will become manly, shows a complete lack of knowledge about the way our bodies work, how hormones play their role and, frankly, it's misogynistic. Women's health would greatly benefit if they stopped doing endless cardio, stopped lifting pretty pink dumbbells to look 'toned' and started doing ass-to-grass squats with a loaded barbell on their back!

Case Study

Dolan Acharya wasn't always this fit.

She started off like so many other women do, with postpartum weight gain. The daily stress became too much and she found solace in food. Then, the inevitable happened—binge eating followed by crash dieting followed by even more binge eating. When Dolan became a part of the Fittr community, she learned the importance of weight training. And again, like so many women, she was aghast! 'What, me lift?'

But look at her today. She has come a long way from being an overweight, under-confident young woman. And let me tell you, she's a beast in the gym.

Would you call her masculine? Not at all, look at those strong curves! She has strength and vitality and has even competed in India's first All-Natural Physique Contest where she picked up a few trophies in the bikini round.

I wish I had more space to cover all the myths, but I think you get the picture.

What follows in these pages is science. This book can change your life, your health and the way you look at the world around you. It will help you become confident, fit and live your best life. So, forget what celebrities and their overpaid fitness 'experts' have to say. It's time to let science do the talking!

Dolan Acharya



Before

65 kg

After



42 kg

NUTRITION

Food=Fuel

Can you fill diesel in a petrol car and expect it to run? Of course not! Your car needs just that one specific type of fuel; otherwise, it simply won't run.

Your body, on the other hand, is a smarter and more complex system. It knows how to convert various types of food into energy so that it can carry out different functions and keep you in good health. Your job, as the owner of the body you live in, is to make sure that you provide the right kinds of food in the required quantities and at the time that your body needs them.

The food sources that your body needs can be broadly divided into two categories: macronutrients and micronutrients.

What Are Macronutrients?

Macronutrients are those foods that your body requires in large quantities, such as grams or kilos. The six macronutrients are carbohydrates, fats, protein, water, fibre and alcohol (of course, alcohol isn't a 'nutrient' and I don't advocate its consumption, and certainly not in macro quantities!).

As we saw in the previous chapter, carbohydrates, fats and protein provide the body with energy. To understand how much energy is supplied by food and specifically these macronutrients, we use the Atwater system, named after Wilbur Olin Atwater.

Atwater proposed a method to measure the amount of heat generated from fats, carbohydrates and proteins. While different foods have different ranges, for ease of understanding, the values were averaged out, and today, we understand that carbohydrates provide 4 calories per gram, fats provide 9 calories per gram and protein provides 4 calories per gram. However, do keep in mind that these are approximations.

A calorie is a measure of energy. The small calorie or gram calorie (usually denoted as cal) is the amount of heat energy needed to raise the temperature of one gram of water by one degree. Kilocalorie or Kcal and Calorie with a capital C are essentially the same things. People often get confused between calories and Calories. 1 Kilocalorie=1 Calorie=1000 calories: the difference is just a small c and a capital C.

Water and fibre are zero-calorie items and don't provide your body with any energy, but they do perform critical functions that keep you in good health.

Later on, we'll look at each macronutrient in detail and understand their respective function and importance in the body. We'll also look at the macronutrient ratios which form the basis of any diet or nutrition plan.

What Are Micronutrients?

Micronutrients are those nutrients that your body requires in small quantities, such as milligrams or micrograms.

If macronutrients are like the fuel that makes your car run, then think of micronutrients as the engine oil, brake oil and grease that keep the various components of the car working properly and that need to be replenished from time to time. In a nutshell, micronutrients help your body maintain overall good health. Micronutrients are broadly of two types: vitamins and minerals.

It is essential to ensure that the food you consume contains a variety of micronutrients as deficiencies can lead to imbalances in your body. We'll get into the details in a bit.

Macronutrients 101

1. Carbohydrates

Calories per gram	:	4
Primary function	:	Energy/Fuel for the body
Chemical formula	:	$(CH_2O)_n$
Food sources	:	Rice, bread, potatoes, wheat, fruits, vegetables, lentils and pulses

My friend, Shekhar, is an investment banker. He has headed meetings that have involved hard negotiations, sparred with investors and cracked some incredibly complex deals. He's tough, smart as a whip, and manages to remain unfazed and level-headed even in the most nerve-racking situations. Yet, he's scared of sugar!

And he's not alone. I know people who jump out of their skin if you so much as show them a teaspoon of the sweet stuff. If someone were to even ask them how much sugar they put in their morning cuppa, they'll go white in the face and cover their mugs with both hands, as if they're scared of being poisoned by those tiny white crystals. For many years, people have been told that sugar is terrible, that it will kill them, make them fat, and will send them to an early grave where they'll pay for the sin of daring to eat sugar.

Sadly, this fear isn't restricted to sugar alone. Rice, bread, wheat, potatoes, all of them have been demonized and labelled dangerous. Dieticians everywhere tell their clients that if they want to lose weight and get fit, then they must stop eating sugar and rice and anything white. Instagram influencers have launched a crusade against these foods and warn their loyal followers that the key to good health and lasting happiness is to banish these foods forever.

These 'evil foods' all fall into the same category: carbohydrates. But what these dieticians and influencers don't tell you is that without carbohydrates, your body would lose out on a critically vital macronutrient which is a storehouse of energy.

What are carbohydrates?

Pick up any of your favourite foods. Chances are, they contain carbohydrates. Whether it's that glass of orange juice you love first thing in the morning or the dal-roti-sabzi that your wife packed for lunch, they all contain carbohydrates. Your idlis and dosas, fried rice, aloo parathas, burgers and pizzas—all contain carbohydrates. Even the fruit chaat that you eat when you're fasting contains carbs.

Carbs are ubiquitous, and yet, most of us don't know them very well. It's time we changed that. For the longest time, carbohydrates have been classified as simple and complex, depending on their molecular structure.

Simple carbohydrates are those which are made up of very few molecules of sugar. Monosaccharides such as dextrose, galactose and fructose have a single molecule/unit of sugar while disaccharides such as sucrose, maltose and lactose have two molecules/units of sugar. Examples of disaccharides are table sugar and fruit juices. Then, there are trisaccharides and oligosaccharides as well.

Complex carbohydrates or polysaccharides are essentially starches and fibres and are composed of multiple sugar molecules. Examples of these are rice, wheat, legumes and vegetables.

Simple carbohydrates are easily and quickly broken down into glucose and enter the bloodstream almost immediately, which is why they're considered a quick and readily available form of energy.

Complex carbohydrates take a longer time to be ingested and don't get transferred directly into the bloodstream. They get broken down into glucose over a more extended period.

Glycaemic Index and Glycaemic Load

If we're talking types of carbohydrates, then let's also discuss two other methods of rating carbohydrates.

Glycaemic Index (GI) refers to how quickly carbohydrates raise the glucose levels in our bodies. Foods such as rice and juice break down faster into glucose, which is why they're considered to be high GI. Glycaemic load rates carbohydrates based on GI and the amount of carbohydrates contained in certain foods.

People tend to simplify (confuse) the issue a tad too much: simple carbs and high GI foods are bad, complex carbs or foods that have a lower glycaemic load are good. That's just not true. Each type of carbohydrate has its place in a healthy diet. To understand this, you need to know what happens to carbohydrates when they enter your body.

Our bodies are made up of trillions of cells. These cells need energy to carry out their vital functions. By far, the best and most efficient source of energy is glucose. Where does this glucose come from? It is derived from carbohydrates. Glucose travels quickly from cell to cell, providing energy. There are two main reservoirs of glucose, active and passive. In its active or ever-present form, glucose is found in the bloodstream, hence the term blood glucose or blood sugar. So when the doctors tell you that your blood sugar is high or low, they're essentially referring to this. In healthy adults, the blood glucose range is between 70–110 milligrams per decilitre, which roughly translates to about 3–5 g of sugar in about 5 litres of blood. As your blood has a limited capacity to store glucose, your body needs to store it elsewhere in the body and retrieve it as and when required. This reserve of the passive form of glucose is called glycogen, and it is stored in your liver and muscles.

Glucose is like your salary, while glycogen is like a bank deposit. You know that your salary is limited, whereas your expenses only seem to increase, no matter what you try. That's why you periodically save a part of your salary in a deposit. Your salary is (or at least, should be) sufficient to meet your daily living expenses, while unexpected or large expenditures can be met from the deposit.

Similarly, your body continually needs energy, even when you're not working: after all, it needs to ensure that your brain, heart, lungs, kidneys and other organs continue functioning. This energy is derived from blood glucose, but just about 4 to 5 grams can be stored in the bloodstream at any given time. That's why your body maintains a reserve of glycogen which can be quickly converted back into glucose when the blood glucose runs out. Between your liver and muscles, your body can store about 100–120 grams of glycogen which can be utilised whenever needed.

Now, what does this tell us? Carbohydrates and glucose aren't bad; they are simply a source of energy. As we saw in the previous chapter, 1 gram of carbohydrates yields 4 calories. As long as you watch your overall calorie intake and don't consume excess calories, you don't need to worry about carbohydrates causing weight gain, since our bodies have a very smart way of regulating how glucose is managed in the body.

Insulin and glucagon

Our bodies have an organ called the pancreas. The role of the pancreas is to secrete two particular hormones called insulin and glucagon. The job of these two hormones is to regulate the blood sugar level and make sure that it stays within the normal range. For an average person, 70–110 mg/dL of glucose in the blood is normal. Given the limited capacity of the blood to store glucose, anything higher than that is considered too high.

Think of insulin and glucagon as highly efficient buddy cops. As long as things are moving along smoothly, they won't interfere. But at the first sign that help is needed, these two policemen reach on time and do their respective jobs.

Let's say you eat a candy bar which contains 30 grams of sugar. As soon as you've eaten it, your bloodstream will instantly receive a hefty dose of glucose, and your blood sugar levels will rise above normal. This is a signal to the beta cells of the pancreas that it needs to secrete insulin.

Insulin immediately starts cleaning up the glucose in the bloodstream and tells the cells of the body that they need to absorb this excess glucose. Some cells use glucose as energy, and whatever isn't needed at that moment is converted into glycogen and stored in the liver and muscles. As the cells take in glucose, the blood sugar levels drop and return to normal.

Glucagon is needed when the blood glucose levels drop below the normal range. This drop happens when you stay hungry and don't consume any food for some time. As levels drop, the pancreas releases glucagon which tells the liver and muscles that it's time to change the stored glycogen into glucose and release it into the bloodstream. This restores the blood sugar levels to normal.

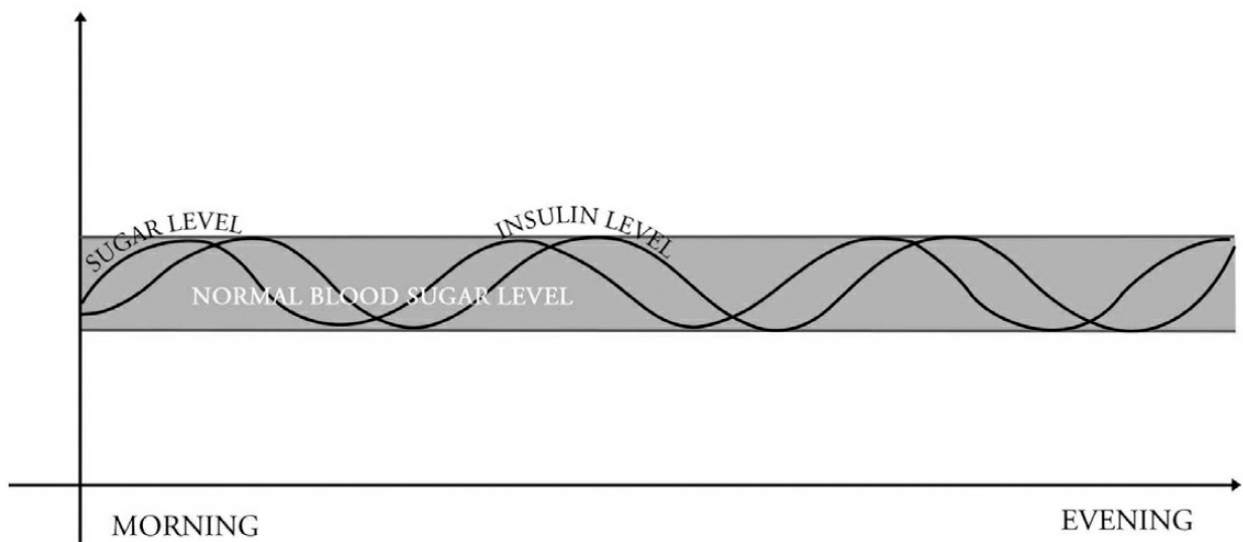
Carbohydrates and hunger

One of the biggest reasons that carbohydrates are demonized is because of the claim that the more carbs you eat, the hungrier you get, resulting in overeating and weight gain. Let's understand this claim.

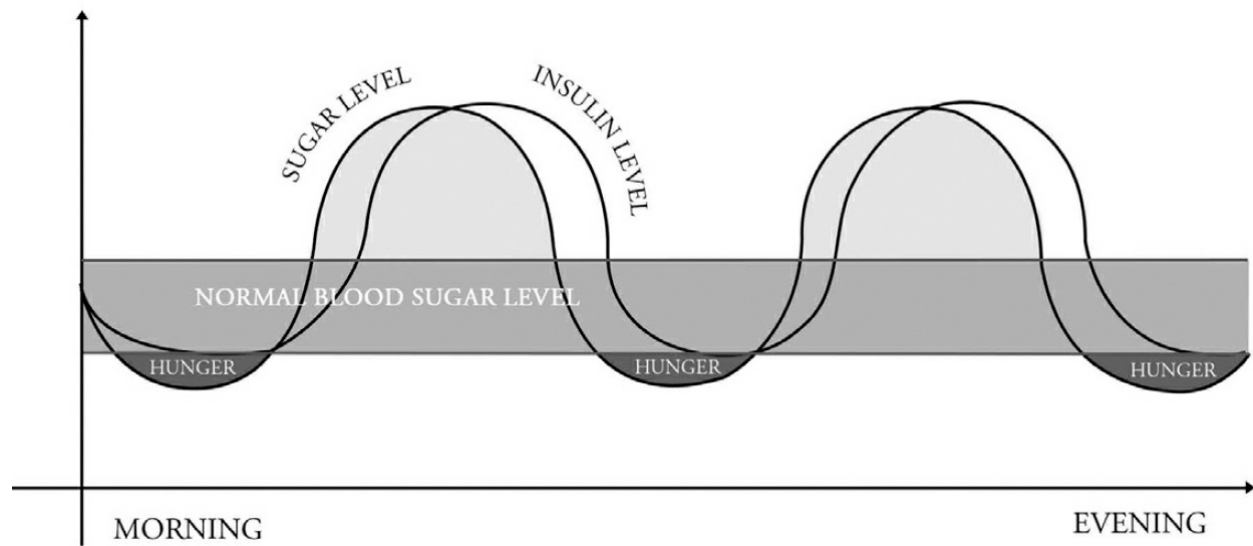
As we've seen earlier, 90–110 mg/dL or roughly 4–5 grams of sugar in the bloodstream is considered normal. Through the action of insulin and glucagon, the body will ensure that blood sugar levels don't shoot up too much (a state called hyperglycaemia) or fall too low (resulting in hypoglycaemia).

In some people, this constant spike in blood sugar and the resultant release of insulin and glucagon leads to a distressed or overworked pancreas. What happens then is that the pancreas starts malfunctioning. When blood sugar levels spike, it doesn't release sufficient insulin or releases it too late, by which time blood sugar has risen excessively.

Healthy adults



Adults with impaired pancreas



Then, to compensate for the delayed response, the pancreas ends up secreting too much insulin which virtually removes all the sugar from the bloodstream. As a result, the body's blood sugar levels don't just drop, they crash.

Carbohydrate-Insulin model of obesity

Dr David Ludwig proposed that it is your intake of carbohydrates and not your overall calorie intake that is to blame for obesity. According to this hypothesis, eating a high carbohydrate diet which includes large amounts of starchy processed foods and sugars leads to accumulation of fat in your cells and makes you fat. This claim does not have any merit. Research has clearly shown that when you keep total calories and protein levels constant and vary the number of carbohydrates and fats you consume, a low carb diet is not better than a high carb diet. Going low carb does not lead to more fat loss just as eating more carbs does not lead to weight gain: overall calories matter!

An often misunderstood fact about insulin is that it is dose-dependent, which means that the insulin spike is proportional to the size of the meal or the amount of carbs. Naturally, if your meals contain fewer carbohydrates, then the spikes would be smaller and take less time to go back to the baseline. But if the meals have more carbohydrates, then the spikes would be larger and would take longer to return to the baseline.

However, the problem isn't with the carbohydrates; even protein can cause insulin spikes. The problem is eating a lot. As long as you're eating food within your TDEE, the spikes shouldn't bother you. Insulin is just doing its job. The notion that eating more frequent smaller meals regulate insulin is as wrong as the idea that suggests eating two big meals will fix insulin. Like I said earlier, insulin is dose-dependent, and what matters is how much you eat, not when. A well-planned quantified diet that contains all macronutrients in the right amounts would ensure hormone regulation.

2. Fats/lipids

Calories per gram	:	9
Primary function	:	Storage reservoir of energy, absorption of fat-soluble vitamins
Chemical formula	:	$\text{CH}_3(\text{CH}_2)_n\text{COOH}$
Food sources	:	Coconut oil, ghee, butter, olive oil, nuts, egg yolks

I'm not a very social guy, but I do attend weddings from time to time. I like the festivities, the joyful atmosphere and the idea of two people coming together. But what I enjoy the most is lurking around the buffet table and spying on people as they navigate the dinner spread. Trust me, if you ever want to see bro-science at play, then there's no better place to eavesdrop!

The conversation usually goes like this:

Lady A: Arrey Sarla, how are you? Long time! You're looking so thin!

Lady B: Haan, no? I'm trying out this new diet. I've stopped eating all fats: no ghee, no butter, no cheese.

Lady A: Then what do you eat? Give some tips, na.

Lady B: See, it's very simple. Fat makes you fat! Stop eating fat, and you'll stop being fat. I'm only eating boiled vegetables and soups. I've even stopped eating parathas for breakfast, they're so fattening! Just two digestive biscuits and half a cup of tea for me.

Lady A: I should also try this. Bhaiyya, paneer mat dena. Sirf gravy do!

However, unlike in this conversation, nowadays there's a whole new trend that has surfaced. Fat has gone from being demonized to being worshipped as the elixir of life itself. Everyone is all about the fat! People have ditched their toast and have started to add butter to their morning coffee. People can't stop talking about coconut rice and ghee and how they contain fat-burning properties. Avocados, a South American fruit that was unheard of in our part of the world, has popped up in gourmet stores and people are making a beeline for it despite its exorbitant price. The reason: you've guessed it—the high fat content.

I wouldn't be surprised if I ran into those ladies today and saw them ditch the rice and fill up their plates with just paneer. So, what has suddenly caused this craze for all things fat? For that, we need to understand why fat was demonized in the first place.

The big fat lie

In January 1961, a physiologist named Ancel Keys was featured on the cover of *Time*. In it, he made a sensational claim that would rock the medical establishment: Americans were killing themselves because they were consuming a high-fat diet. These were the findings of the so-called ‘Seven Countries Study’ which Keys had launched in 1958. According to his results, there was a direct link between consuming dietary fat and the prevalence of coronary heart disease.

Almost immediately, the American Heart Association declared that a low-fat, low-cholesterol diet was the best bet if someone wanted to keep heart disease at bay. The message was clear: the villain was fat, and it had to be banished from the diet. They recommended two things: first, that carbohydrates in the American diet be increased from under 50 per cent to over 70 per cent of daily calories. Second, at the same time, reduce the quantity of fat consumed. Meat and fat were replaced with starches and cereals. Everything sold in supermarkets became ‘low-fat’, including the butter. This demonization of dietary fat went on unabated for decades.

But just like fashion, fat too has made a comeback with a lot of help from the low carb movement. Keto, Paleo, LCHF, call them by any name you want, these zealous advocates of the high fat-low carb diet will tell you how fat has been wronged, and we’re all worse off because of it. Apparently, fat doesn’t make you fat; carbohydrates make you fat! Go figure.

So, who’s right? What should a person who wants to live a healthy life do? The answer, as always, isn’t black and white.

Lipids

When we talk about fats, what we're talking about are lipids. Lipid is a generic term which includes those substances in the body which are insoluble and cannot be transported via the bloodstream. These include triglycerides, cholesterol and phospholipids.

Triglycerides are the most common type of lipids and are found in the body as well as in the food we eat. Triglycerides are stored in adipose tissue, which is the main storage location in the body. When someone says that they want to lose fat, they're talking about triglycerides, which is the stored form of fat in the body. Just as carbohydrates are converted into glucose, triglycerides in the food we eat get converted into fatty acids, and they are a source of energy for the body as well. Whenever you consume any food, the fats in the food get broken down into free fatty acids and glycerol. Free fatty acids yield energy (just like glucose), and are utilised by the body to carry out various functions. But if they don't get used, then the free fatty acids combine with a molecule of glycerol and form triglycerides.

Here's the unique thing about triglycerides: they can be stored anywhere in your body. And, if there aren't enough cells to store all the fatty acids, then your body is quite capable of generating brand new fat cells (called adipocytes) to accommodate these fatty acids. There is no limit to how many adipocytes or adipose tissue your body can create. In theory, this means that there is no limit to the amount of fat your body can store. (Scary thought, huh?)

Essential fatty acids: Omega 3 and Omega 6

There are several different types of fatty acids: Omega 3, Omega 6, Omega 7 and even Omega 9. Of these, your body can produce Omega 7 and Omega 9 fatty acids. Since our bodies are capable of synthesizing these fatty acids, they are called 'non-essential' fatty acids.

It then follows that essential fatty acids are those which cannot be synthesized in the body and must, therefore, be obtained from the foods we eat. That's why Omega 3 and Omega 6 fatty acids are called 'essential' fatty acids.

Saturated versus unsaturated fats

Just as there is a raging debate about fats versus carbohydrates, there's a separate debate about saturated and unsaturated fats. Whenever ads tell us which healthy oils make healthy people, they generally recommend unsaturated fats, also known as refined vegetable oils.

Saturated fats are fatty acids with a single carbon bond, while unsaturated fats are fatty acids with double carbon bonds. Unsaturated fats are further classified into monounsaturated and polyunsaturated. Monounsaturated fats or MUFAs are those that contain a single double bond. When more than one such double bond exists, they're called polyunsaturated fats or PUFAs.

Saturated fats such as butter, coconut oil and ghee tend to be solid at room temperature. Hence, the argument goes, you should consume them sparingly as they can clog the arteries! Other foods that contain saturated fats include bacon, cheese, meats and whole-fat cream: all the devils of the low fat movement that are venerated by the low carb/keto crowd.

Yet again, the truth gets lost in the noise.

Cholesterol: Is it really bad?

Another common accusation levelled against saturated fats is that they raise cholesterol and cause heart disease. Let's examine this.

Without getting too technical, cholesterol can be described as the most potent antioxidant that is at your body's disposal. It is a fundamental component of every cell membrane in the human body, including brain and nerve cells. Cholesterol enables the production of Vitamin D from sunlight and protects brain cells against the ravages of free radicals. It is a precursor to the production of the sex hormones: estrogen, progesterone and testosterone. Cholesterol is also essential for the production of hormones released by the adrenal glands, such as cortisol, corticosterone, aldosterone etc. Cholesterol aids in the production of bile, the digestion of fats and plays a vital role in the metabolism of fat-soluble vitamins (A, D, E and K). It also prevents ion leaks in membranes and is essential for insulin release.

The human body is extremely smart. Given how vital cholesterol is, it does not rely on food to complete the requirements of cholesterol. Instead, it produces cholesterol on its own. If we do consume cholesterol through food, our bodies are smart enough to regulate the amount of cholesterol it makes.

Cholesterol is considered to be the cause behind heart conditions such as ischemia and atherosclerosis. When doctors examined patients for arterial blockages and atherosclerosis, they found a substance that formed a plaque in the arteries and caused clogging. This substance was cholesterol. In a classic case of correlation being mistaken for causation, cholesterol was blamed for heart disease. The fact is that it is not cholesterol that is the culprit but rather, the oxidized carriers of cholesterol, known as high-density lipoprotein (HDL), low-density lipoprotein (LDL), very low-density lipoprotein (VLDL), triglycerides and chylomicrons. These carriers transport cholesterol to various places in the body where it is needed.

In some cases, the carriers get damaged due to oxidation and cholesterol starts leaking into the blood. As it is insoluble in the blood, it keeps floating around until it is cleared from the blood by the liver. But often, this leaking cholesterol forms a plaque which then becomes a problem.

The solution to this problem is not to lower the cholesterol but to reduce the factors that cause the oxidation of its carriers. These factors include smoking, drinking, the presence of free radicals caused by stress, pollution and glycation, and a constantly elevated blood sugar level. Addressing these issues is the key to good heart health, not reducing cholesterol.

The blood test for reviewing cholesterol levels is, in reality, a count of its carriers.

HDL particles remove fat molecules from cells which need to export fat molecules. The fats carried include cholesterol, phospholipids and triglycerides.

LDL delivers cholesterol to cells (most importantly, brain cells), where it is used in membranes, as an antioxidant or for the synthesis of steroid hormones. LDL is our friend. One should try not to decrease LDL.

VLDL transports endogenous triglycerides, phospholipids, cholesterol and cholesteryl esters. It

also serves for the long-range transport of hydrophobic intercellular messengers, like the morphogen Indian hedgehog (protein).

TGL (Triglycerides) are stored in your fat cells. Later, hormones release triglycerides for energy between meals.

Chylomicrons serve to transport fat from its port of entry in the intestine to the liver and adipose (fat) tissue.

3. Protein

Calories per gram	:	4
Primary function	:	Building muscle, repair and growth
Chemical formula	:	$RCH(NH_2)COOH$
Food sources	:	Eggs, poultry, meat, soya, paneer, legumes, dairy

Protein comes from the Greek word ‘proteos’ which means ‘of utmost value’. This should tell you what an important role this macronutrient must have in building your health. Proteins are complex forms of amino acids. An amino acid is a simple organic compound comprising a carboxyl group ($-COOH$) and an amino group ($-NH_2$).

Unlike the other two macronutrients we’ve seen, the primary function of protein is not energy production. Protein has been called the building block of health, and it really is so. Muscle growth, the quality of your hair, nails and skin, and overall development of the body, all these would be adversely affected if your protein intake is insufficient. And if you’re an Indian reading this book, chances are very high that your protein intake is woefully inadequate.

Essential and non-essential amino acids

Protein is made up of amino acids. When you consume any food containing protein, it gets broken down by the action of digestive enzymes into its component amino acids.

There are nine essential amino acids and 11 non-essential amino acids. Essential amino acids need to be obtained from the food we eat, since our bodies can't synthesize these by themselves. So, the quality of food as a source of protein is judged by its amino acid profile—the more complete its profile, the better. Some foods which are complete sources of protein are eggs, chicken and meat. Among vegetarian sources, only soya is a complete source of protein.

THE 11 NON- ESSENTIAL AMINO ACIDS

Alanine
Arginine
Asparagine
Aspartic Acid
Cysteine
Glutamic Acid
Glutamine
Glycine
Proline
Serine
Tyrosine

THE 9 ESSENTIAL AMINO ACIDS

Histidine
Isoleucine
Leucine
Lysine
Methionine
Phenylalanine
Threonine
Tryptophan
Valine

Vegetarian and non-vegetarian protein

If we're talking about protein, we need to address the elephant in the room—yes, non-vegetarian sources like the ones listed above are superior sources of protein. Just deal with it.

Vegetarian sources are considered inferior because they don't always contain all nine essential amino acids, a fact that is undoubtedly going to raise the hackles of my plant-eating friends who are reading this.

I also mentioned earlier that Indians are deficient in protein, and the fact that our population is mostly vegetarian is certainly a contributing factor. However, there is a way for vegetarians to ensure that even they get to consume all nine essential amino acids, through a simple trick called 'smart food combinations'. Here's how it works: simply choose vegetarian food sources that are complementary to each other and make up for each other's amino acid deficiencies. Combine them in the same meal so that you end up getting all nine essential amino acids.

For example, rice contains protein, but it lacks the amino acid lysine. At the same time, beans contain lysine, but they lack the amino acid methionine which rice already has. So, if you were to eat rice and beans together, you would end up consuming a meal with a complete amino acid profile. Simple, isn't it?

Case study

Anupam Adarsh Mishra is a fitness coach with Fittr. Would you believe me if I told you that he got this fantastic physique on a purely vegetarian diet?

Anupam was scrawny—the kind of person who was always teased and bullied for being thin. He wanted a body worthy of being on the cover of a fitness magazine, but as a student living on a tight budget in a hostel, he had no idea if it could be done. But, he found a way. With the help of Fittr, he educated himself about nutrition and realized that even as a vegetarian, he had plenty of food options that could make him bulk up and become fit. Anupam followed a very simple diet: soya, rice, dal and paneer. That's it! He didn't even eat eggs! And yet, look at the results.

Anupam Mishra



Before

50 kg

After



70 kg

How good is your protein?

Apart from its amino acid profile, there are other ways to judge how good a protein source is.

Biological Value (BV): Did you know that eggs are called the gold standard of protein? That's because eggs have a BV of 100. BV is a measure of how efficiently the body can utilise the protein that it absorbs from food. In other words, it measures how much nitrogen is absorbed from the food you eat.

Net Protein Utilisation (NPU): This takes into account how much nitrogen is utilized/retained once the body absorbs it. If nitrogen excretion is high, that means NPU is high.

Protein Digestibility Corrected Amino Acid Score (PDCAA): This is what the FDA currently uses for measuring the quality of protein. While other methods tell you the overall quality of protein, PDCAA looks at the digestibility of the individual amino acids, which makes it a much more comprehensive measure.

Protein and muscle growth: muscle protein synthesis

A combination of nutrition, exercise and adequate rest is needed to ensure that your muscles see proper growth.

When you work out or do any physical activity, your muscles experience wear and tear. The muscle fibres literally break down, which is known as muscle proteolysis. These muscles need to be repaired. Muscle protein synthesis refers to how the body repairs and grows muscle tissue after exercise.

What would happen if your expenses exceeded your income? You'd be in the red, right? How would you survive? In the same way, if your muscle proteolysis or breakdown is more than your muscle protein synthesis, then your body will not be able to repair and grow the muscle tissues, resulting in loss of muscle mass. However, if muscle protein synthesis is more than muscle proteolysis, then your body will be able to repair the broken down muscle tissue and also grow them in size.

Proper protein intake can decrease muscle protein breakdown, which is why it is essential to make sure you consume an adequate amount of good quality protein. So, exactly how much protein do we need to consume?

Values vary from 0.8 grams per kg of body weight all the way to 2.8 or even 3 grams per kg of bodyweight. That's a pretty wide range!

My recommendation: decide for yourself. Start with the recommended dietary allowance (RDA) of 0.8 grams of protein per kg of body weight. Keeping this as the minimum threshold, consume as much protein as you can. Check your tolerance for higher amounts of protein. As long as you are not experiencing digestive distress or flatulence, you're good to go! If you do feel uneasy, reduce your protein consumption.

There are some apparent benefits to consuming higher amounts of protein: first, it will help ensure that muscle protein synthesis is more than muscle proteolysis (which is very important, as we've seen earlier). Second, it has also been found that higher protein intake bumps up your metabolic rate, sometimes by 18 to 20 per cent. Now, who wouldn't want that? So, find your own protein sweet spot.

Optimal amount of protein per meal

I've been asked this question in almost every forum: is it true that you should not have more than 30 grams of protein per meal? If I eat more than that, will that protein go to waste? Worse yet, will it be converted into fat?

Now that you know the concept of energy balance, I hope I don't need to explain to you that no food group is solely responsible for fat gain. Overall calories matter.

Research does show that as far as muscle protein synthesis is concerned, about 40 gm of protein per meal might offer the maximal response. However, it's still not clear whether this is a one-size-fits-all recommendation.

Again, just as you did with the amount of protein you consume in a day, find your protein-per-meal sweet spot. The important point is to make sure that you meet your daily protein requirement.

Protein and women

If I had a dollar for every time I've heard that protein makes women manly, I would have enough money to buy a lifetime's supply of protein. These 'experts' are also the same people who tell women not to lift heavy weights at the gym (more on that later in the book).

Newsflash: women have muscles, too! And just like men, their muscles also need help to repair and grow. Ladies, don't listen to these knuckleheads. Please go have your protein!

Protein and aging

Another myth is that older people don't need protein since their physical activity levels have declined. Nothing could be further from the truth!

As we age, our muscles start to atrophy. Age-related sarcopenia is a widespread condition where the amount, quality and strength of the skeletal muscle mass begins to deteriorate over time. It becomes imperative to consume an adequate amount of protein even at that age so that our bodies don't break down.

While it is true that our overall calorie requirements reduce as we age, the component of protein in an ageing diet needs to be kept high.

4. Water

Calories per gram	:	0
Primary function	:	Maintaining core body temperature, keeping the body hydrated, maintaining blood levels, absorption of nutrients
Chemical formula	:	H ₂ O

Yes, water is a macronutrient and yet, people drink such micro quantities of it! Jokes aside, water is vitally important. You can survive for a few weeks without food, but only three to four days without water, which makes it all the more alarming that so many people walk around in a state of dehydration.

Water makes up nearly 60–70 per cent of our bodies. Almost all our organs are made of varying amounts of water: blood, in fact, is nearly 90 per cent water with the rest being plasma.

How much water should we drink?

When we say that we must drink water, we need to take into account all fluids that we consume. Tea, juices and even the so-called 'dehydrating' coffee contains water and counts towards our daily water intake. Even the fruits and vegetables and other foods we eat contain water and count towards our fluid intake. However, to be on the safe side, we only consider fluids. And no, fluids do not include alcohol! Let's not use any excuse to justify drinking.

Fluid intake guidelines

General recommendations: 3.7 litres per day (men) and 2.7 litres per day (women)

By body weight: 1 litre of fluids for every 23 kg of bodyweight

By thirst: If you're thirsty, drink water (not a great way to measure)

By urine colour: On a scale of one to eight, urine colour can guide you about your state of hydration or dehydration. Seven or higher means you're dehydrated and your urine looks a very dark shade of yellow, but I'd not suggest waiting that long before drinking water.

Other factors such as your level of physical activity, climate, frequency of urination, all these would ultimately determine how much water you need.

Effects of low water intake

Low water intake affects:

- the kidneys and prevents them from performing optimally
- the health of joints, since water helps keep them lubricated
- your metabolism adversely
- your digestion adversely
- the quality of your skin, hair and nails

Myths about water

While we're on the subject of water, there are some wildly outrageous lies that have appeared on social media and even in respectable newspapers which need to be addressed. Ridiculous as they are, I'm worried that there are people who believe them.

Myth #1: Don't stand up when you drink water: According to this myth, the pull of gravity on the water you drink can damage your organs and even hurt your knee joints.

First, water doesn't just fall into your stomach. Whenever you eat or drink something, it travels through the food pipe which isn't a hollow tube but rather, a collection of muscles which expand and contract to allow the passage of the food and drink. Whether you stand up or sit down, the position and placement of the food pipe will not change, and water will travel through it in precisely the same manner.

Second, the question of water hurting your knees doesn't arise. There isn't even a passage that can directly take water from your mouth to the knees. Water doesn't drop like a rock onto your knees: this notion is too ridiculous for words.

Myth #2: Drinking water while eating dilutes stomach acids and causes indigestion. Your stomach acid is a combination of hydrochloric acid, potassium chloride and sodium chloride. The pH of this mixture is around 2, which is pretty high, and would require you to drink almost 2 litres of water immediately to bring the pH up to about 3.5 which is still acidic enough. Your stomach meanwhile will continue secreting more bile to maintain the pH, so it's challenging, if not impossible, to dilute your stomach acid by drinking water in between your meals.

5. Fibre

Calories per gram	:	0
Primary function	:	Provide bulk to stools and promote gut health

The last macronutrient we will look at is dietary fibre. Like water, it provides no calories or energy but is vital for good health, especially gut health. There are two main types of fibre: soluble and insoluble.

Soluble fibre

Soluble fibre, as the name suggests, gets dissolved in the body. It forms a thick gel-like substance and swells up. Good sources of soluble fibre include grains, fruits, vegetables, legumes, etc.

Soluble fibre is not absorbed in the body; instead, it is fermented in our colon and promotes the growth of good gut bacteria. It also makes us feel full and can suppress appetite, which is why soluble fibre is essential on a diet. They also produce short-chain fatty acids which have several benefits for the body.

Psyllium husk—better known in India as Isabgol—is a common supplement that is consumed as an aid to digestion and evacuation.

Insoluble fibre

Insoluble fibre, on the other hand, doesn't dissolve. It retains much of its original form as it passes through our intestines, carrying waste material out of the body. Sources of insoluble fibre are the husk of wheat, bran of grains, etc.

By aiding the evacuation of waste from the body, insoluble fibre promotes gastrointestinal health, such as reducing the risk of haemorrhoids and constipation.

Dosage

If your diet is rich in green leafy vegetables and you also eat a regular serving of fruits and vegetables, then you won't need to supplement dietary fibre. Your body does well with a daily dose of about 25–30 grams of dietary fibre, whether from food or supplements such as Isabgol. Avoid falling into the trap of high fibre biscuits—they're a scam. Almost every cereal you eat regularly has enough fibre, apart from vegetables and fruits.

Micronutrients 101

Although our bodies need micronutrients in small quantities, in grams and milligrams, don't make the mistake of thinking that they also play a small role in your overall health. Micronutrients are critical for good health, and you need to guard against deficiencies.

Types of Micronutrients

Micronutrients fall into two broad categories, inorganic minerals and organic vitamins.

Minerals are further classified into two types: macrominerals, or minerals that our bodies need in relatively larger quantities, and trace minerals, or those needed in very small quantities. Macrominerals include calcium, phosphorus, potassium, sulphur, sodium, chlorine and magnesium. These are electrolytes and have a critical role to play in fluid intake and balance. Trace minerals include iron, copper, zinc, cobalt, molybdenum, iodine, selenium, etc. These play a role in maintaining our physiology.

Vitamins are also of two types: fat-soluble vitamins such as A, D, E and K and water-soluble vitamins such as Vitamin C and all eight forms of Vitamin B. Fat-soluble vitamins are absorbed into our bodies with the help of lipids. They tend to accumulate in the body, so you need to be careful if you're supplementing them. On the other hand, the interesting thing about water-soluble vitamins is that it's difficult to overdose on them. Since they are water-soluble, they get flushed out of the body, so they must be replenished daily.

Benefits of micronutrients

Micronutrients play vital roles in the growth and repair of the body. They also aid the production of hormones. This is the reason why the body regulates minerals very tightly. However, our bodies are entirely dependent on external dietary sources for obtaining the required doses of vitamins.

Foods such as fruits and vegetables, eggs and meat are the best sources of vitamins and minerals. Dark green leafy vegetables are, hands down, the best sources of micronutrients.

To mitigate the risk of deficiency, it may be a good idea to supplement with a daily multivitamin. I have covered this at length in the chapter on supplements. Always note that supplements are not replacements: you should always try to obtain your macros and micros from foods first.

Hormones

A discussion on nutrition would be incomplete without a look at hormones. Hormones play a significant role in everything that your body goes through.

Hormones are special chemical messengers that pass messages between different parts of your body and control various vital functions in your body. They coordinate complex processes like growth, fertility and metabolism. Do you have facial hair? Hormones. Do you have good muscles naturally? Hormones. You feel very energetic? Hormones. From your menstruation to your thyroid functions, everything is controlled by hormones. Are you diabetic? Hormones, again.

1. Insulin

Insulin has always been propagated as the muscle-building hormone, as well as the fat-storing hormone. The truth is, it does both, and you don't need to fret much about it. It's a hormone that helps you stay stable, by keeping your blood glucose levels stable when it tends to rise due to incoming carb intake. It 'opens up' the cells to take in the glucose.

Insulin is a hormone made by the pancreas which allows your body to use sugar (glucose), from carbohydrates in the food that you eat, for energy or to store glucose for future use. Insulin prevents your blood sugar level from climbing too high (hyperglycemia) or falling too low (hypoglycemia).

The cells in your body need sugar for energy. However, sugar cannot enter most of your cells directly. After you eat and your blood sugar level rises, cells in your pancreas (known as beta cells) are signalled to release insulin into your bloodstream. The insulin then attaches to and signals cells to absorb sugar from the bloodstream. Insulin is often described as a 'key' which unlocks cells, to allow sugar to enter them and be used for energy.

2. Glucagon

We discussed how insulin helps keep the blood glucose level stable by assisting the body to take in the excess glucose. Glucagon does the exact opposite; it helps the blood glucose remain stable by breaking down the stored glycogen in the liver and bringing in a supply of glucose when blood glucose is low. So, insulin and glucagon are interlinked, and are produced in the pancreas.

3. Testosterone

Testosterone is the primary male hormone and gives men their 'male' characteristics, from voice to facial hair. It helps maintain proper health in men and also helps to build muscles. Even women produce testosterone, but in lesser quantities than men, and are more sensitive to it. A healthy testosterone level is vital for good muscle gain and fat loss. So, make sure your micronutrients are on point and that you are not on a hypocaloric diet for long. (There is a difference between calorie deficit and hypocaloric: the latter means severe deficit for a long time.) The best testosterone booster in nature is lifting heavy weights. Lifting weights improves

your testosterone production which, in turn, aids muscle building and strength gain. In women, ‘toning’ is nothing but muscle building, along with fat loss. Lifting weights will help you do that instead of aerobics.

4. Estrogen

Estrogen is the female hormone and is responsible for the development and regulation of the female reproductive system and secondary sex characteristics. Now, there are several different types of estrogen, but that is beyond the scope of this book. One thing to remember here is that it is a critical hormone for women, and lifting weights will *not* reduce it to low enough levels to affect your health. Another concern among men is about the increase of estrogen in their bodies due to soya consumption: that won’t happen unless you eat 1 kg of raw soya every day continuously for months at a stretch.

5. Leptin

Leptin is an essential hormone in your body that aids in fat loss. It is like the body’s fuel indicator which signals your brain about the availability of food. Based on that, your brain holds on to your body’s fat stores, or goes easy on them to release them for use. This is directly related to metabolic adaptation—the way your body controls and varies your metabolism. This hormone falls when you are on a caloric deficit for too long, and this effect is greater if carbs are omitted from the diet.

6. Ghrelin

This hormone is also known as the hunger hormone. It signals your body to feed it more. The levels of this hormone go up when you are on a caloric deficit for long and are dieting. So, the next time you feel hungry, don’t think that your body lacks energy or energy stores. It has a lot of fat stores that it can use: it’s just the ghrelin in action.

7. Cortisol

In the fitness community, cortisol is seen as a villain. The truth is, it is neutral, but it depends on how it functions. It is a catabolic hormone that helps in the metabolism of fats, protein and carbohydrates. Even your fat loss is dependent on cortisol. However, excess amounts of cortisol for extended periods is harmful, as it can metabolize your muscle mass too. Cortisol is released in response to stress and low blood glucose concentration.

This was a long chapter, and I’m glad you’ve stuck with me right till the end. In the next chapter, we will finally come to a very vital discussion: how to prepare your own nutrition chart.

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PREPARE YOUR OWN NUTRITION PLAN

Introduction to Quantified Nutrition

I've been talking about quantified nutrition for some time now, but before we look at what it is, let's first understand what it is not. Quantified nutrition is neither about throwing together an arbitrary bunch of foods just because they feel right and look good together, nor is it about eating certain foods just because that's what your grandparents ate or because they are 'locally sourced'.

Quantified nutrition is also *not* about excluding foods. People make the mistake of labeling foods as 'good' or 'bad': carbs are evil, but fats are healthy, so you can eat as much butter and ghee as you want. White rice is terrible, but brown rice is good. Ditto for white versus brown bread, sugar versus jaggery or honey, and milk versus soy milk, and so many other foods. Instead, quantified nutrition is about including a variety of food items as long as they build your overall health and help you achieve your fitness goals.

Dieting isn't a jamboree where everyone is invited, it's a well-planned event where nutrition science is the bouncer, whose task is to let in the right foods through the door. A proper quantified nutrition chart will, therefore, include a variety of foods but in the most optimal quantities.

So, what is quantified nutrition? As the name suggests, quantified nutrition is a type of diet strategy where you need to *quantify* or *measure* the food that you eat. When you plan your nutrition or diet plan, you have to create a list of the food items you will consume and their quantities. In a nutshell, this is quantified nutrition.

Now, I don't know about you, but a lot of people become worried when they hear this. (I think it's because mathematics wasn't their strongest subject in school.) 'So, you mean that I have to measure every single thing that I eat?'

Just the thought of doing this makes people want to give up entirely. To add fuel to the fire, some 'dieticians' and 'wellness experts' scoff at the idea of measuring food and feed this fear. They tell their social media followers that they pity the fools who quantify their food. They want you to believe that people who follow a diet are prisoners of their food choices. They make it sound as if counting calories is so complicated that people are wasting their time and energy doing it. The most absurd statement of this sort was one that asked people to count their blessings and not calories. Let me get this straight: you want people to count something that

cannot be quantified but not something that can be easily measured? And surprisingly, people believe that!

But here's the truth: quantified nutrition isn't difficult, it's actually quite simple.

How does quantified nutrition work?

My wife is an excellent chef. Sadly, the cooking gene has wholly bypassed me. What I do love is to watch her cook. She's an absolute wizard in the kitchen; there isn't anything she can't cook. Somehow, she knows exactly how much spice and salt to add to every dish. She's so good at it that she just has to catch a whiff of the dish that's cooking, and she knows what's missing.

'This needs a little more salt,' or 'Just a pinch of garam masala, and it's done.' I've heard her say these so many times and she always gets it right. Even before she starts cooking, she knows exactly how much rice or dal or flour she needs to use to prepare the meal. Measuring food is almost second nature to her and she could do it with her eyes closed. She knows that if she needs to make five chapatis, then she needs 3 cups of flour, a pinch of salt, 2 teaspoons of oil and about 3 cups of lukewarm water. And once the dough is kneaded, she can divide it into five pieces, each of the same size. How about some rice and dal for the pressure cooker? Out comes that cup again to measure out the required quantities. How does she do it? Practise, practise, and some more practise.

If you think about it, what my wife is doing is quantified nutrition. In fact, every single wife, mother, grandmother and chef since the dawn of time has been following the principles of quantified nutrition. All we have to do is to take that instinct for measuring food and give it a more systematic approach. What's wrong with the unsystematic approach? Well, try measuring your waist with your hands and give those measurements to a tailor and you'd know. I will be showing you different ways to follow quantified nutrition and achieve your fitness goals. These will make it so much easier to measure your macros and calories and stick to your diet.

Goals and Key Numbers

There are two things you need to do before you sit down to prepare a diet plan that works: one, set your fitness goal and two, fix your numbers (calories and macros). Once you have these details in place, you can easily prepare your own nutrition plan.

You've all heard of SMART goals: Specific, Measurable, Achievable, Realistic and Time-bound. Usually, I loathe management jargon, but this term is an exception. Let's see how you can apply this principle to set your fitness goal.

1. Specific

The first rule of setting a goal: be specific. 'I want to lose weight' is vague. 'I want to lose fat' is much better.

Our bodies are made up of muscles, bones, tissues, organs, water and fat. When you step on a weighing scale, the number you see is a sum of all these things. If your goal is to lose weight, the next question is: exactly which part of your total weight do you want to lose? You obviously can't lose your organs; you need them. You also don't want to lose your precious muscle. So, the part of your weight that you want to shed is your fat.

When Pavan came to me, he had already been working out for over two years, and he had made decent progress. He had an okay physique but wanted to take it to the next level. I asked him, 'What specifically do you want?' Pat came the reply, 'I want to reduce my fat percentage to below 10 per cent.'

See, Pavan had learned the hard way that being wishy-washy about your fitness goals will also get you sub-par results. Being specific about what you want is the first step. Ask and you shall receive, right? And let me tell you, Pavan achieved his goal with the help of a quantified diet that was tailor-made for his specific goal.

Here's a good goal: 'I want to lose 10 kg of fat.' It's a good place to start, but it's still an incomplete goal. There are other factors you need to consider, as well. Read on.

2. Measurable

How will you measure that you are losing fat? I know what you'll say: 'I will weigh myself every day. The days on which the scales show a dip, I've lost weight (sorry, fat) and on days when the scales show a rise, I'll collapse into a crying heap in the corner.' Sounds like a brilliant strategy for losing your mind.

The scales are deceptive. A better strategy is to check your progress with a measuring tape: inch loss always trumps weight loss.

Here's how it works: maintain a diary where you keep a record of your body measurements. With the help of a measuring tape, note down:

- waist circumference
- hip circumference
- arm/bicep circumference

- belly circumference (i.e. measure your stomach across your navel)

Track these measurements week after week and check if you are making progress over time. In addition to that, you can also keep checking your weight, but use this smart way of doing it: starting on a Sunday, weigh yourself daily first thing in the morning at the same time. Do this after visiting the loo and before eating or drinking anything. Keep track of your daily weigh-ins. The next Sunday, calculate the average of the previous week's weigh-ins. Track this week after week to track your progress.

3. Achievable

I keep seeing this request from a lot of people, especially women: 'Please sir, help me lose 10 kilos in four weeks. My best friend is getting married, and I want to look my best!' I honestly don't know if I should laugh or feel sorry for them. The truth is that you did not gain so much fat overnight. How can you expect to lose it overnight? Men, on the other hand, usually want to put on muscle and get six-pack abs in the shortest amount of time possible. Also not going to happen. In reality, muscle gain is much more difficult than fat loss.

So, set a goal that can be achieved. Chasing a goal that doesn't have a hope in hell of seeing the light of day will only lead to frustration and disappointment.

4. Realistic

This is related to the previous point about setting an achievable target. Is your fitness goal within reach? Is it realistic and relevant to your life?

Let's face it: if you're 5'4" tall and you want to look like Hrithik Roshan, that is simply not going to happen, or at least not in this lifetime. Now, I may have been guilty of hero worship in the past myself—and there is nothing wrong with holding someone as an ideal or inspiration—but you need to be realistic.

Find what you can and cannot do with what you have. Be inspired, don't blindly copy.

5. Time-bound

It's good to set a goal that has a start date and an end date. Of course, only a goal that is realistic and achievable can be given a deadline. At the same time, don't be daunted by the path that lies ahead of you.

Let's say you're at 40 per cent body fat, have a 38-inch waistline and are pre-diabetic. You will need to give yourself a lot of time to get fit. But if you break up this mega goal into smaller mini-goals, then you will see results and stay the course.

Can you reduce your waistline from 38 inches to 34 inches in, say, 12 weeks? Or perhaps aim to halve or even get off insulin within three to six months? (Yes, diabetes is not a life sentence!) Success with these smaller goals will eventually give you the confidence to chase bigger results.

Fix your numbers (BMR, TDEE and calories)

Now that you have fixed your SMART fitness goal, it's time to prepare your diet plan. But first, let's do a recap:

Calories are a measure of energy. When you eat food, it provides your body with a certain amount of energy which is measured in calories.

When you eat 1 gram of protein, it yields 4 calories of energy.

When you eat 1 gram of fat, it yields 9 calories of energy.

When you eat 1 gram of carbohydrates, it yields 4 calories of energy.

Macros refer to the quantity of each macronutrient (protein, fat and carbs) that you choose to include in your diet. So, when someone asks, 'what are your protein macros?' what they want to know is how many grams of protein you have included in your diet plan.

Understanding Maintenance Calories

To refresh your memory:

- $\text{Calories in} < \text{Calories out} = \text{Fat Loss}$
- $\text{Calories in} > \text{Calories out} = \text{Weight Gain}$
- $\text{Calories in} = \text{Calories out} = \text{Maintenance}$

In short, maintenance calories are the calories at which you neither lose weight nor gain weight. You maintain your current weight. This means that if you want to lose fat, then you need to eat below maintenance calories, and if you want to gain weight, then you need to eat above maintenance calories.

How to find your maintenance calories

To know your maintenance calories, you need to calculate your BMR and TDEE. There are three ways to find out these numbers.

1. Use an online calculator

This is the easiest and quickest method. Any online BMR and TDEE calculator would do. The Fittr app has an excellent one, where all you have to do is enter a few details, and the calculator does the rest.

2. Use good old algebra (ugh!)

If you're mathematically inclined, then the second method should work.

Step 1: Calculate your BMR

The Harris-Benedict formula is most commonly used to calculate BMR:

Women's BMR = $655 + (9.6 \times \text{weight in kg}) + (1.8 \times \text{height in cm}) - (4.7 \times \text{age})$

Men's BMR = $66 + (13.7 \times \text{weight in kg}) + (5 \times \text{height in cm}) - (6.8 \times \text{age})$

Step 2: Determine your activity multiplier

Activity Multipliers	
1.2	Sedentary; little or no exercise
1.375	Lightly Active; light exercise 1-3 days per week
1.55	Moderately Active; moderate exercise 3-5 days per week
1.725	Very Active; hard exercise 6-7 days per week
1.9	Extremely Active; hard daily exercise

Step 3: Calculate your TDEE

$\text{TDEE} = \text{BMR} \times \text{activity multiplier}$

For example, let's say that your BMR is 1500 calories and you have a sedentary lifestyle. You get some exercise, but for the rest of the day, you're hacking away at your computer. Your activity multiplier would hence be 1.2. Thus, $\text{TDEE} = 1500 \times 1.2 = 1800$

3. Use trial and error

There's a third method, but that needs some observation and effort on your part.

For a couple of weeks, track your food intake and body weight every single day. You need to be meticulous. Maintain a food log where you write down the exact foods and the quantities of each that you consume throughout the day. This practice will also give you a very clear picture of what you eat all day, including any unconscious snacking that might be going on. Along with this, check your bodyweight every single morning as we've discussed earlier in this chapter.

After doing this for a fortnight, check if your weight has fluctuated. If your weight hasn't budged, congratulations: you've found your maintenance calories. If you've put on weight, then you're most likely eating above your maintenance calories and need to reduce your calorie intake. The reverse is true if you've lost weight.

This method is excellent because it gives you a very good idea of what food you're consuming and how your body is responding to it. But it is complicated and prone to fail. You will need to track every morsel and sip that you take throughout the day for several days and let's face it, you're probably not going to get it right. That is why I recommend using any of the other two methods.

Now that you know your maintenance calories, you can start creating your nutrition plan.

Create Your Own Nutrition Plan

This is it, folks! The moment you've all been waiting for. It's time to create your own personal nutrition plan and take charge of your health. I know this is new and uncharted territory for many of you, so let's take it step by step.

NUTRITION PLAN		
PART A – BASIC INFORMATION		
Personal Details		
Name:		
Age:		
Height:		
Weight:		
My Fitness Goal:		
Key Details		
Step 1 — Fix Your Daily Calories		
My BMR:		
My TDEE:		
My Daily Calories:		
(For Fat Loss: more than BMR but less than TDEE)		
(For Building Muscle: More than TDEE)		
(For Maintenance: Equal to TDEE)		
Step 2 — Set Your Macros		
Protein:		
Fats:		
Carbohydrates:		
Step 3 - Select The Right Foods (add as many as you like)		
Protein:		
Sr. No.	Name of Protein Source	Calories (per 100 gm)
Fats:		
Sr. No.	Name of Fars Source	Calories (per 100 gm)
Carbohydrates:		
Sr. No.	Name of Carbohydrates Source	Calories (per 100 gm)

If we're talking about a plan, then it has to be written down. That's how you achieve it. To make things super easy, I've designed a simple-to-use nutrition plan template.

This is similar to the template that our clients at Fittr receive when they enrol with a coach, but since you're doing this all by yourself, I've added a few more points. The template is in two parts: Part A contains all the basic information needed for creating a nutrition plan and Part B is the actual nutrition plan. All you have to do is follow each step and fill in the template, and by the end of this section, you'll have your very own nutrition plan!

Part A: Start by writing your name, age, height and weight. Then, write down your goal. I find that writing things down helps me get them done. Now, let's start filling the rest of Part A.

Step 1: Fix your daily calories

How many calories you need to consume will be entirely dictated by your fitness goal. For muscle gain, you will need to consume calories that are **more** than your TDEE. For fat loss, you will need to consume calories that are **less** than your TDEE. In other words, you need to create a caloric deficit. Generally, I recommend a deficit of not more than 200–300 calories per day. This, coupled with exercise, will create a sufficient impetus for fat loss. Under no circumstances should your calorie intake fall below your BMR. As we've noted earlier, doing so would interfere with the normal processes of the body, which is highly undesirable.

Let's look at an example:

Say, your BMR is 1,400 and TDEE is 1,800. At a deficit of 200 calories, you would need to consume 1,600 calories. You can consider going down to 1,500 calories as well, but that is too close to your BMR so I would advise against it.

In the template: Find out your BMR and TDEE using any of the methods mentioned earlier and add them to the sheet. Then, add your desired/ideal calories.

Step 2: Set your macros (protein, carbohydrates and fats)

Now that you have decided how many calories to consume, you need to choose how many of those calories will come from protein, carbohydrates and fats, so you need to set your macros.

Always set your protein macros first: based on numerous research studies on the topic of optimal protein intake, we find that the range varies from 0.8 grams/kg of body weight to 3 grams/kg of body weight. In my case, I can easily have 2.4 grams of protein per kg of bodyweight.

There are some recommendations about optimal protein intake that will make the decision easier:

Optimal Daily Protein Intake for Adults (g/kg*)

	Of healthy weight		Overweight	Pregnant	Lactating
Sedentary	1.2 - 1.8		1.2 - 1.5	≥ 1.8	≥ 1.5
Active	1.4 - 2.0	1.4 - 2.4		Unknown	
Goal	Maintenance	Muscle Gain	Fat Loss		

My recommendation is to eat at least 0.8 to 1 gram of protein per kilo of body weight to start with, and then keep bumping it up till you get to 1.6 to 1.8 grams. If you've already been consuming protein in the past, you can directly start from 1.8 grams without any issues. The idea is to allow your gut/stomach to adjust to new protein intake, and it's a gradual process.

Step 3 - Select The Right Foods (add as many as you like)
 Protein:

Sr. No.	Name of Protein Source	Calories (per 100 gm)

 Fats:

Sr. No.	Name of Fats Source	Calories (per 100 gm)

 Carbohydrates:

Sr. No.	Name of Carbohydrates Source	Calories (per 100 gm)

From the remaining calories, set your carbohydrate and fat intake. Here's the simple rule: let taste and preference be your guide. Remember that a diet is not a sprint but a marathon. You need to give yourself time to accomplish your fitness goal, so you will need to create a nutrition plan that you can follow for a reasonable amount of time without getting bored and without

giving in to food cravings.

The easiest way to fall off a diet is to exclude the foods you enjoy. The more enjoyable a diet is, the more sustainable it will be. If you love your rice, dal and potatoes, then, by all means, include more carbohydrates in your diet. Your tastebuds love them, so why deprive them? But if you prefer a more low carb approach to fitness, then first set your fat macros and then set your carbohydrate macros from the remaining calories. The guidelines for carbohydrates and fats are easy to remember: make sure you include at least 1 gram per kg of carbohydrates and 0.5 gram per kg of fats in your diet. Keep these minimum recommended intakes in mind when you are designing your nutrition chart.

Let us look at an example to make things clear. At 1,600 calories and bodyweight of 77 kg, you decide to have 2 grams of protein. Thus, your protein intake is set at 154 grams (or $154 \times 4 = 616$ calories). This means that you have 984 calories (1600-616) to fit in your carbohydrate and fat intakes. If you're a fellow potato lover, then you too will favour carbohydrates over fats. So you decide to keep your carbohydrate intake at 150 grams per day (or $150 \times 4 = 600$ calories). That means you have 384 calories to fit your fats (1600-616-600). At 9 calories per gram, that means you can have roughly 43 grams of fats.

Thus, using this approach, your macros will look something like this:

Total calories	:	1600
Carbohydrates	:	150 grams
Protein	:	154 grams
Fats	:	43 grams

In the template: Fix your protein, carbohydrate and fat macros and add them to the sheet. How much protein are you planning to consume? How many carbs and fats? Write them down. Check the figures. Do they match?

Step 3: Select the right foods

Adherence to a nutrition plan improves when you eat the right foods. By this, I don't mean that you have to select some 'superfoods' or practise 'clean eating'.

Here are a few guidelines for choosing the right foods:

1. You like their taste and don't mind eating them every day

Everyone has certain flavours and mouthfeels that they prefer over others. My wife loves the ripeness of tomatoes and adds them practically to everything she eats. I like the taste of butter but can't stand coconut oil. I also love potatoes and can eat them morning, noon and night.

A diet does involve some sacrifices, but there's no reason why it should feel like a punishment. My suggestion is to make a list of three sources each of protein, carbohydrates and fats that you enjoy, and prepare meals using those ingredients. Here's what my list would look like:

Protein	:	paneer, soya and rajma
Carbohydrates	:	rice, potatoes and lentils
Fats	:	butter, ghee, olive oil

Tell me that you can't come up with recipes that use your favourite foods and that are diet-friendly. I dare you!

2. They are convenient to make, carry and store

Meal prep can literally save your diet. It saves you time during the week, so you don't have that excuse for not following your diet. Plus, you will always have food stored in the fridge, so you won't be tempted to order in after coming home from work.

Choose foods that can be prepared in advance, which can be stored for a few days in the fridge without loss of flavour and texture and that you can carry in tiffin boxes to the office or your friend's home when you just want to chill but don't want to eat anything that will throw you off your diet.

3. They are easy to find and simple to replace/swap with other foods

This serves two purposes: one, you don't have to waste time, energy and money on getting them, and two, it can add a little variety to your diet.

Rice can be replaced with oats, eggs can be swapped with cheese cubes (Did you know that the macros of 1 egg and 1 cheese cube are quite similar?) and one type of lentil can be swapped with any other.

What will you do if you've decided to use almond flour as a staple in your diet? Out of pocket and out of food options isn't a great way to be on a diet. Make your life a little easier.

In the template: Make a list of all the food items that you love. Categorize them into proteins, carbohydrates and fats. Once you have the list ready (and I hope it's a long and varied one), also note down their calories. If you can't find any particular food item, a simple online search will give you the answer. The Fittr app also has an extensive food database with thousands of food items with all the calorie and macro details you need.

Congratulations! You are now inches away from creating your nutrition plan. Now, let's move to Part B of the nutrition plan template.

PART B – YOUR PERSONAL NUTRITION PLAN

Food Item	Grams	Calories	Protein	Fats	Carbohydrate
Meal 1: Breakfast					
Sub-total (i)					
Meal 2: Snack 1					
Sub-total (ii)					
Meal 3: Lunch					
Sub-total (iii)					
Meal 4: Snack 2					
Sub-total (iv)					
Meal 5: Dinner					
Sub-total (v)					
TOTAL (i)+(ii)+ (iii)+(iv)+(v)					

Step 4: Divide your macros into meals

Now that you know about calories and macronutrients, it's time to sit down and plan your meals. A lot of things have been said about meal frequency and meal timings. I'm going to touch upon a few of these statements before I tell you that they're unimportant.

- No, you don't need to eat 6 meals a day.
- No, you don't have to eat every 2 hours.
- No, you don't need to eat only 2 meals a day, each lasting no more than 51 minutes, and then consume nothing throughout the day except water or green tea.
- No, you don't have to fast for 16 hours a day and consume all your meals in an 8-hour feeding window.

If you *want* to do any of these things, feel free to do so. But no, you do not *have* to do any of them. Frequency and timing may matter to pro athletes but the average person—you and I—need not worry about these factors.

The only things that matter to us are: (a) paying attention to the energy balance (calories in vs calories out) and (b) completing our daily macros, especially our required protein intake. It would help to pay attention to your specific daily schedule and accordingly plan when and what to eat. For instance, I'm not a big breakfast guy and usually have my first meal around 1 p.m. But I know people who need to eat something as soon as they wake up, which is perfectly fine. Similarly, some people may decide to skip dinner or just have something light while others can't go to bed on an empty stomach. Find what works for you and work with that.

In the template: The template is already divided into five meals: breakfast, lunch, dinner and two snacks. You can use the same format or change things according to your schedule. Maybe you just want to eat three meals a day? That's perfectly fine. The important thing is to make sure that you complete your macros for the day. A number of meals will make it convenient to manage your energy throughout the day. You know your daily schedule better than anyone else. Find what works for you.

Step 5: Some important do's and don'ts

You don't need to be a Nigella Lawson or Jamie Oliver in the kitchen to make your diet food edible. Anyone can cook a simple, nutritious meal. And when in doubt, check out the food recipes on the Fittr app.

Do: Invest in some basic tools

They'll make your life so much easier, and they don't cost a lot of money.

- kitchen weighing scale which costs about ₹300
- a good quality non-stick frying pan and kadhai
- measuring spoons (Recipes that call for ½ cup and ¼ spoonful are aplenty, but if you decide

to use your household katoris or spoons to measure the ingredients, you're bound to get the measurements wrong. The difference between 1 measured tablespoon of oil and what you think 1 tablespoon looks like could easily put you over your daily calories.)

- Storage containers to keep your prepped meals in the fridge/freezer

Do: Measure your foods uncooked

Whatever you're eating—rice, chicken, dal, potatoes, paneer, oats—always measure the foods dry and uncooked. Cooking alters the volume of almost every food, and you won't know what quantities you're consuming. So, if your diet includes 100 grams of rice, that's 100 grams of uncooked rice. Similarly, 200 grams of chicken refers to raw chicken.

Don't: Count the calories of the following ingredients

dry spices such as cumin, turmeric, oregano and other dry seasonings and condiments

green leafy vegetables

fibre (Some people differ on this point. Technically, fibre is a carbohydrate, and so it does contain some calories, maybe about 2 calories per gram. However, fibre doesn't get digested but rather, aids elimination by adding bulk to faecal matter. And what doesn't get digested doesn't provide calories.)

Do: Count the calories of the following ingredients

saucers and other wet condiments/flavourings (soy sauce, salad dressings, mayonnaise all contain calories and, in some cases, quite a few)

vegetables, especially starchy vegetables (some people say that it's okay to ignore the calories from vegetables. But if you're having three to five servings of vegetables per day, which is a good idea, by the way, then those calories can add up quite fast.)

Step 6: Start planning your meals

You know your calories and macros, you've made a list of which food items you wish to eat, you have your tools in place, and the food is quantified. Now comes the next step: start adding food items to each meal.

To make things easier, here's a diet chart that I had prepared for a friend. He's on a 2,000 calorie-plan and is a vegetarian. A busy professional who works long hours, he needs energy throughout the day.

SAMPLE NUTRITION PLAN – VEG (2000 calories)

Food Item	Grams	Calories	Protein	Fats	Carbohydrate
Meal 1: Breakfast					
Butter/Oil/Ghee	5	36	0	4	0
Paneer	75	209	14	17	2
Soya Chunks	45	156	24	0	15
Sub-total (i)		401	38	21	17
Meal 2: Snack 1					
Soya Chunks	45	156	24	0	15
Sub-total (ii)		156	24	0	15
Meal 3: Lunch					
Butter/Oil/Ghee	5	36	0	4	0
Spinach	150	0	0	0	0
Paneer	100	282	18	22	3
Rice	65	225	4	0	51
Sub-total (iii)		543	22	26	54
Meal 4: Dinner					
Rajma	60	200	14	0	36
Rice	80	277	5	0	63
Butter/Oil/Ghee	5	36	0	4	0
Soya Flour	85	346	30	18	15
Sub-total (iv)		859	49	22	114
TOTAL (i)+(ii)+(iii)+(iv)		1959	133	69	200

Do you see how simple this is? Go on, try it. Prepare your own nutrition plan!

Common Problems Faced by Quantified Dieters and Their Solutions

I get it, life is busy. You have jobs and responsibilities and bills that need to be paid. Kids and parents, pets and potted plants that take up all your time. Then, you have friends and family who want to invite you to dinner, bosses who want you to take clients out to lunch, plus weekends that are designed for fun and not boring things like a diet.

We live in fast times and have hectic lives that demand all our energy and attention. We can barely catch our breath, where do we have the time to measure food? Indeed, there's an easier way, a quick fix—a magic wand that you can wave or a pill that you can take—and hey presto, you're thin!

Sadly, there isn't. Calories in versus calories out is the only way to lose fat, gain muscle and stay in good health. Sure, you can throw the kitchen scale out and guesstimate how much you're eating. But sooner than later, this will fail. You'll end up eating too much or too little, and when you don't achieve your fitness goal, you'll be left wondering why your diet hasn't worked.

Quantified nutrition isn't complicated, but it might seem a bit daunting at first. That's why I thought of putting together a few tips that will make it easier for you to get started. In my experience, once you get over these initial hiccups, you'll become a quantified nutrition convert for life!

Problem: Cooking for yourself when you also have to cook for the whole family

Solution: This can be a challenge because you may not have the time to prepare meals in two separate batches, one for yourself and one for the rest of the family. To solve this issue, you have a couple of options.

Prepare everyone's meals together and measure your portion of cooked food: When you cook food, it either loses water or soaks it up. For instance, when you cook rice and dal, the water you use adds to the volume and weight of the cooked dish. On the other hand, foods like chicken lose moisture and shrink in volume and weight. That's why you must have noticed that a small cup of raw rice becomes a big bowl of cooked rice. 60 g of raw white rice can increase in volume and weight to become 150–180 g of cooked white rice, whereas 100 g of raw chicken breast could weigh less after cooking.

Depending on how you prepare food (boiled, steamed, fried or sauteed), the volume and weight of the cooked dish changes. When you cook food in a large batch, it can become difficult to know how much of that food you need to eat, as per your diet plan. Check the food database in the Fittr app which also has the calories and macros details of cooked food. This will help you to stick to your macros and ensure that you don't go off your diet.

Use the same measuring bowls and spoons every time: While this technique can work, it's best to measure your food before cooking. If you find it tedious to do this every time that you cook, you can put aside some bowls and spoons that you use for measuring food. Better yet, purchase some measuring cups and spoons (1 cup, ½ cup, ¼ cup and 1 tablespoon, 1 teaspoon, ½ teaspoon and ¼ teaspoon). All you have to do is use the kitchen weighing scale once to measure how

many grams of each food item fits into each cup and spoon.

For instance, 1 cup of raw rice is approximately 200 g, while 1 cup of wheat flour is about 120 g. Do this once for each of the food items in your diet chart and then use the same measuring bowl and spoons every time you cook. You can use the same technique for measuring cooked food as well. Just remember, measure 1 **level** cup/spoon and not 1 **heaped** cup/spoon.

Problem: No time to cook

Solution: Cooking meals in advance can be a lifesaver or, in this case, a time-saver. There are several advantages, including the fact that you don't have to slog in the kitchen every single day. Besides, whenever you want to eat, you have cooked food in the fridge that you simply have to heat and eat—no reaching for the food delivery app at the end of a long day.

I suggest cooking meals for three to four days at a time and storing them in the fridge. On Sunday, you could cook all your meals for Sunday to Wednesday and pop them in the fridge. And on Wednesday, cook your meals for Wednesday to Saturday. Cooked food stays fresh in the fridge for up to four to five days (longer, if stored in the deep freezer). For measuring the food items, use any of the techniques given above.

Problem: I'm a frequent traveller. How do I quantify my meals in hotels?

Solution: This is a tricky problem which calls for some creative solutions.

Carry home-cooked meals for shorter trips: If you can eat a dabba at office, you can do it for out-of-town trips as well. Of course, if you're travelling for more than a day, do this only if you have a fridge in your hotel room.

Packaged foods: Ready-to-eat meals come with complete nutrition information on the pack. You get everything from poha and upma, to biryani and sabjis. All you need is hot water or a microwave, both of which are found in any hotel.

Protein bars: Keep these handy and also carry a few scoops of whey protein powder with you when you travel.

Boiled eggs: When in doubt, reach for boiled eggs. You don't need to measure them, just count them.

Carry your measuring cups and spoons: You will turn heads when you pull these out at the dining table, but who cares? You're on a diet, you have a fitness goal and you need to stick to them.

Eyeball your food: This technique is a little hit-and-miss and should be used carefully. Since you've been measuring or quantifying your food at home, you can guesstimate how much you're eating.

Choose foods wisely: Skip the dal tadka and have plain dal. No fried rice for you, stick to plain rice. Let your friends and colleagues eat that oily Mughlai chicken, you will have grilled chicken and paneer tikka, thanks! And don't forget to help yourself to salad (without the sauces and dressings).

Problem: How can I follow a quantified diet and still have a social life?

Solutions: Here are some solutions for this specific and common problem.

Plan ahead: If you know you have a party to attend, eat about 100 calories less, starting a few days leading up to the event. You'll have a few spare calories at your disposal.

One serving only: Don't reach for second helpings. If possible, serve yourself—friends and relatives tend to go overboard when they serve you food (level spoonfuls, not heaped spoonfuls, remember).

Be that weird friend who carries their measuring cups and spoons (Just kidding...or am I?)

Problem: Is there a way to follow a quantified diet without using a kitchen weighing scale?

Solution: Frankly, this isn't recommended. If you've been following a quantified diet for some time, you can try and eyeball your food from time to time, but the chances of overeating will always be there.

Even if you are using measuring cups and spoons, you will need to use the kitchen weighing scale at least once to check the exact weights. Don't be afraid of the kitchen weighing scale or think of it as an inconvenience, it is an indispensable tool and will help you achieve your fitness goals.

Problem: Can vegetarians follow a quantified diet? How do we get our protein?

Solution: There are many members of the Fittr community who are vegetarians or vegans and who have still lost weight, gained muscle and look incredible.

It's a myth that vegetarians will never get adequate protein. All you need to do is use food smartly. The issue with vegetarian sources of protein is that most of them don't contain all nine essential amino acids. But if you combine the right foods in the same meal, you can solve this issue to a great extent.

Problem: Quantified nutrition sounds expensive

Solution: This is another myth. When you follow quantified nutrition, you're cooking your own meals, which will always be cheaper than ordering or eating out. Additionally, the benefits of eating quantified home-cooked meals will also make you fitter and healthier over time. This means fewer sick days and fewer trips to the doctor. Money saved!

What would you rather spend money on—good, healthy food or hospital and medicine bills?

Sample nutrition plans

You're just starting your fitness journey, so let me make things a little easier for you. I've already shown you a vegetarian nutrition plan for 2,000 calories.

Here is another sample plan, this time for the non-vegetarians. Feel free to tweak it to your requirement. But remember, these sample plans are a guide, not a crutch. This is personal fitness, so, personalize it. Apply what you've learned so far, prepare your own nutrition plans, create them for your friends and family as well. Trust me, when they start seeing results, they'll thank you.

SAMPLE NUTRITION PLAN – NON-VEG (2000 calories)

Food Item	Grams	Calories	Protein	Fats	Carbohydrate
Meal 1: Breakfast					
Butter/Oil/Ghee	8	58	0	6	0
Whole Eggs	5 nos.	345	30	25	0
Sub-total (i)		403	30	31	0
Meal 2: Snack 1					
Banana	100 gm	95	0	0	24
Sub-total (ii)		95	0	0	24
Meal 3: Lunch					
Butter/Oil/Ghee	8	58	0	6	0
Spinach	100 gm	0	0	0	0
Chicken Breast (Boneless)	150	191	38	5	0
Rice	90	312	6	0	71
Sub-total (iii)		561	44	11	71
Meal 4: Snack 2					
Chicken Breast (Boneless)	100	127	25	3	0
Sub-total (iv)		127	25	3	0
Meal 5: Dinner					
Rajma	40	134	10	0	24
Rice	90	312	6	0	71
Butter/Oil/Ghee	8	58	0	6	0
Chicken Breast (Boneless)	150	191	38	5	0
Sub-total (v)		813	56	13	119
TOTAL (i)+(ii)+(iii)+(iv)+(v)		1999	155	58	214

EXERCISE AND TRAINING

There's an old saying: abs are made in the kitchen. But that's only half the story. Nutrition plays a vital role in making you fit and healthy, but if you want to look good, build strength and live a life that's free of aches and pains, you *have* to exercise.

But where do you even begin? If you thought all those myths about nutrition were bad enough, just wait till you read the misinformation about exercise and workouts:

‘Weightlifting is only for bodybuilders.’

‘You can’t get a body like that without steroids.’

‘You should change your workout every couple of days,
this will confuse your muscles and they’ll grow!’

‘To get a six-pack, do 1,000 sit-ups every day.’

And two of my personal favourites:

‘Women should not lift weights.’

‘Cardio is the best exercise for fat loss.’

It’s not exercise that’s the problem; instead, exercise seems to have a misinformation problem! Ask yourself this question: how many of those exercise myths I just mentioned did you or do you still believe? I’ll bet that you thought at least a couple of them were the gospel truth. And, it’s also possible that some of you sat up and said, ‘Hey! But of course cardio works, all I do is run on the treadmill and I’ve lost so much weight.’

Remember that story I told you about the time I tried to lose weight and not fat? I try not to think of it myself.

Here’s the truth: most people don’t have the slightest clue about which form of exercise they should do. Is walking really the best form of exercise? Should women really not lift weights? Is yoga all you need to do? People seem to have set notions about exercise that refuse to go away. And that actually prevents a lot of people from ever taking that first step and even trying to work out.

Proper nutrition without proper workouts is like trying to drive a car with a full tank but a flat tire. Sure, you’re moving ahead, but eventually, you’ll be forced to stop because the wheels will simply give up. But try telling this to a naive 18-year-old kid that the influencer whom he

follows so ardently on social media doesn't know anything about exercise; that if he were to follow this 'fitness' influencer's instructions, it would do him more harm than good. And it's not just youngsters who don't know any better. I once tried telling my friend's mother that walking to the market twice a day didn't qualify as exercise. (Man, I miss her cooking).

Ask yourself these questions (and answer them honestly):

- Wouldn't you love to wake up looking like a million bucks?
- How about not having to hold your breath to fit into your favourite pair of jeans?
- Don't you wish you didn't have to hide your double chin in your selfies?
- Don't you want to feel full of energy through the day?

Everyone—and I do mean everyone—wants to look good, and there is absolutely nothing wrong with that!

Case Study

Ashish (named changed on request) is a senior executive in a tech company with offices all over the world. He is smart, knowledgeable, and an excellent co-worker and boss. Everyone likes him. When it was announced that a new head of Latin America would soon be appointed, everyone—including Ashish—knew that he would be the best fit for the job. But somebody else got that promotion.

Disheartened, Ashish confided his disappointment with his boss and asked what he had done to deserve this kind of treatment. Here's what his boss told him: 'Ashish, you're a great guy, but have you checked the mirror lately? You look like a heart attack waiting to happen! We can't have someone like that engaging with clients in such an important geography.'

In short, Ashish lost his promotion because he was fat!

I'm sure that's illegal, and Ashish might win a lawsuit, should he choose to sue. But his case isn't a unique one. People make snap judgements about you all the time, based solely on the way you look. Whether they're looking for a new head of business or a life partner, people are almost always guided by how you look. Some of you might retort that beauty is only skin deep and what matters is who you are on the inside, but the world doesn't work that way!

Evolution has primed human brains to seek out people who look good, healthy and strong. Our brains assume that a person who meets these physical attributes is better, more capable and therefore, more deserving of respect than someone lower down the scale. You might think I'm being harsh here, but I've been there. Ask yourself, what's the most recent film you watched? Who was the lead? A good looking, handsome, tall guy? What do your kids want to be like when they grow up? A strong superhero or a beautiful princess?

Nobody wants to be fat or unattractive. When you are fit, you feel confident. You have energy, vitality and strength. You engage with life, you don't shy away from social occasions. What's more, you stop standing behind plants, pets, your kids, billowing skirts and large handbags when you have your photograph taken, hoping against hope that no one will see your not-exactly-petite waistline.

Exercise is about more than just looking good. It also has several other health benefits:

Builds strength and bone health: Regular exercise builds a stronger body. It's good for your heart, your lungs, your muscles and bones, and general overall health.

Prevents age-related ailments and muscle loss: As you grow older, it becomes more important to protect the health of your bones, joints, and muscles. If you're physically inactive, you open yourself up to numerous health issues in your advanced years such as the risk of fractures, the risk of falling, arthritis and other rheumatic conditions. Your muscles grow weaker and even performing everyday activities such as sitting down and getting up become a problem. Exercise can mitigate these issues and even prevent them from cropping up in the first place.

Builds up immunity in the long run: Stop turning to pills and powders to 'boost' your immunity; the best way to do that is by indulging in regular exercise. This happens in a number of different ways—exercise increases blood flow, lowers inflammation and stress, strengthens

antibodies, and impacts the white blood cells which help your body fight diseases and infections.

Reduces the risk of chronic diseases: Research shows that regular physical activity can reduce your risk of developing chronic illnesses such as cardiovascular disease, type 2 diabetes, and even some cancers.

Bolsters brain health and memory: There's an indirect yet crucial link between exercise and improved cognition. Exercise improves your mood and sleep, while it also reduces stress and anxiety. This has been shown to improve your memory and brain function.

Helps with depression and creates a general sense of well-being: Regular exercise not only gives you a sense of accomplishment but actually releases hormones such as dopamine, serotonin and norepinephrine which can improve your mood and make you feel better and happier.

At Fittr, we love to say that we don't just put years into your life, but we put life into your years. Exercise is that key which helps you create a healthy life, build immunity and freedom from illness, and live an active and energetic life well into your silver years and beyond.

Just as you created your own personal nutrition plan, you also need to have your own exercise or workout regime. But before that, you need to understand the basics. That's what this chapter is all about.

Components of Fitness or Why Do We Exercise?

We could call these the basic reasons why you exercise in the first place. Here's a general rule of thumb to remember: those exercises and workouts that include most or all of these components are better than an exercise plan that 'feels' right but is lacking these necessary components:

Cardiorespiratory Fitness: This is a measure of how well your heart and lungs are able to work together for supplying oxygen and fuel during periods of physical activity.

Strength: Muscles don't just make us look good but also help us perform daily activities without aches and pains. If you have the confidence to move your bodyweight while doing pull-ups or push-ups, you won't shy away from lifting suitcases for fear of throwing your back. One of the most common causes of aging is that your body loses this ability and starts wasting away. The right kind of exercise can help prevent this.

Flexibility: Healthy joints make a healthy body. If your joints are flexible, you will be able to move around without pain. Ask an older person what they would give if only they could once again be able to sit down and get up without help. Good flexibility also prevents injuries because your body can easily perform its full range of physical movements.

Ideal Body Composition: Simply put, lower body fat and higher/adequate lean muscle mass is desirable. Higher fat levels are associated with lifestyle diseases, chronic illnesses and a lower quality of life.

Different Types of Exercise Routines: An Overview

It's also a good idea to learn about the different types of exercise forms and workout protocols that are available. Different goals call for different workouts. At the same time, it's important to know what each type of exercise can and cannot do:

• Weightlifting/Resistance Training:

What It Is: When people think of gym workouts, this is usually what comes to mind—a ripped guy working out using a barbell loaded with plates. And they'd be right. Resistance Training is the form of exercise where you provide *resistance* to your muscles through some kind of *external load* (in this case, the weights). The stress of lifting weights triggers a chemical reaction in your muscles and prompts them to grow. A proper diet, especially one with adequate protein, is non-negotiable.

Suitable for: Building muscle and strength, losing fat.

• Powerlifting:

What It Is: This is a more specialized form of lifting weights. The idea here is to test the limits of your strength and power by progressively lifting more and more weights over time. Powerlifting is concerned with executing the movement while lifting maximum possible weight and with proper form and technique. Olympic lifts fall under this category.

While resistance training is usually more about building a good physique, powerlifting is about building power.

Suitable for: Building power, improving your strength.

• Yoga:

What It Is: Hatha Yoga is an ancient system of physical exercises that was primarily a spiritual practice but has now become a way to build a healthy body and mind. Sun Salutations and Downward Dog have entered the English lexicon, thanks to yoga's popularity in the West.

Using yoga for weight loss is a bit like using a diamond for breaking stones. Yoga is much more than just physical poses—it's a holistic system that uses postures, breath and mindfulness to cultivate calmness and stillness in your being.

Suitable for: Flexibility, mindfulness, integrating body, mind and emotions.

• Bodyweight Exercise/Calisthenics:

What It Is: The COVID-19 pandemic had only one positive outcome—it made people realize the need to start taking better care of their health. Home workouts became—and continue to be—quite popular because, after all, where do you go but your living room when you can't go anywhere else?

Bodyweight training and calisthenics are also a form of resistance training where your own body provides the load. Calisthenics is famous for looking almost like ballet—a beautiful art form that also gives you buns of steel. To make things interesting and to add some additional resistance, you can use resistance bands or external weights such as backpacks filled with bottles or even household items like buckets.

Bodyweight training tends to take a little longer to build your dream physique as compared to lifting weights, but with regular practice and proper exercise progression, it will

happen.

Suitable for: Anytime and anywhere fitness, building muscle, balance.

. **HIIT (High-Intensity Interval Training):**

What It Is: Another variation of the ‘anywhere, anytime’ workout is this extremely sweaty and intense way of getting jacked. Like bodyweight exercises, no equipment is needed. The exercises are done in very rapid succession, with very short to no breaks between them.

Let’s say you did 10 High Jumps, 10 Jumping Jacks, 20 Bodyweight Squats and then ran 100 meters, all without a break—you’ve just done HIIT. To ramp things up, just do more exercises, more repetitions, take shorter breaks and do multiple cycles. Not for the faint of heart but highly satisfying if you want to pump up your heart rate and feel the burn.

Suitable for: Burning fat, heart health, speed and agility.

. **Functional Training:**

What It Is: As we grow older, it can become slightly more challenging to carry out our regular physical movements. Bending, sitting down and getting up, stretching, and balance can get compromised. That’s what functional training aims to retain.

It’s a series of exercises designed to ensure that your body—joints, muscles and bones—remain in peak condition, and don’t develop signs of physical wear and tear.

The exercises simulate actual movements while also developing strength. Typically, the exercises involve some combination of Push, Pull, Squat, Lunge, Hinge, Twist and Walk, which cover all the possible movements of the body. Weights can be used or even just your bodyweight.

Suitable for: Agility, balance, daily movements.

. **Cardio/LISS:** Low-Intensity Steady State cardio is what you must have seen a lot of people do at the gym—it’s when you get onto the treadmill or elliptical or stationary bike, set the speed at a slightly brisk but comfortable pace and then get going. For that matter, LISS is also what you do when you walk in the park or go jogging.

As the name suggests, LISS is a form of cardio where you maintain a steady pace which isn’t too difficult. Your pulse is elevated, but you’re not out of breath.

Suitable for: Heart health

So Which Exercise Program Works for You?

The components of fitness are your destination while the workouts or exercises are the vehicle that will take you from your current state of health to your ultimate destination. That's why it is critical that you choose the right kind of exercise.

Clearly, some exercises or workouts are better than others. But there's one form of exercise that stands head and shoulders above the others—Resistance Training.

It doesn't matter if you're a man or a woman—if you want to be fit and healthy with a body that's the envy of everyone, you *need* to lift weights!

I. INTRODUCING RESISTANCE TRAINING

Resistance Training: As Old As Mankind

Call it by any name—weightlifting, weight training, strength training or resistance training—it's been around for thousands of years.

From the Ancient Greeks to the Ancient Romans to our very own Harappan ancestors, we've always idolized physically strong heroes.

Who's the strongest man in Indian mythology? Someone who could beat his opponents on the battlefield and win the day? Of course, Bhima from the *Mahabharata*.

Who's the strongest God? Well, that's a debatable topic right there. But if we were to talk about physical prowess, then the name that comes to mind is our own Bajrangbali, Hanuman!

What do Bhima and Hanuman have in common with other mythic heroes such as Hercules and Androcles? Their awe-inspiring strength and powerful physiques.

How did they build such bodies? By lifting weights!

What Is Resistance Training?

Simply put, resistance training is a form of exercise where you move your body *against* some kind of resistance. This resistance could come from your own body when you push it away from the floor, or when you're trying to pull yourself up against the gravitational force or—as we commonly understand it—when some external weight such as a barbell or dumbbell or even your own bodyweight is lifted.

All these are types of resistance training. The particular form that involves lifting weights is called weight training.

A Brief History of Weight Training

There's substantial evidence that weightlifting as a sport has been around since the 10th century BC. One of the earliest recorded instances involves China's Zhou Dynasty, which was famed for its large army. All recruits to the army were required to pass weightlifting tests before they were accepted into the military.

But this wasn't anything new. Most ancient tribes around the world had so-called 'Manhood Stones'—huge boulders that were used by young men of the tribe to show off their strength and power. The strongest men were seen as valuable members of the tribe. They were given priority over others in matters such as the use of useful resources and even choosing a mate.

And how did you become strong? The logic was simple—you don't lift heavy objects because you are strong. But you become strong by lifting heavy objects. And the more you lift, the stronger you become.

In modern times, the pioneer of weight training and bodybuilding was Eugene Sandow. He earned that body the old-fashioned way—by lifting heavy. He inspired countless others to start taking care of their bodies which is why they call him the 'Father of Modern Bodybuilding'.

But somehow, over time, a myth has become quite commonplace: weight training is only meant to be done by 'bodybuilders' or by those men who want to get into modelling or win physique contests.

The other major myth is that only men should lift weights—women should stick to cardio and tiny pink dumbbells.

Before going any further, let's bust these two myths.

Myths About Weight Training

Myth #1: Only bodybuilders/models/athletes should lift weights:

The truth—lifting weights offers innumerable benefits:

- **Lean Muscle:** The resistance offered by lifting weights causes microtears to develop in your muscles. Coupled with a proper diet, this stimulates muscle growth and helps you build lean muscle mass.
- **Fat Loss:** Let's also understand the concept of Excess Post-exercise Oxygen Consumption (EPOC) or the 'after-burn' effect. After exercise, your body requires more oxygen to return to its regular resting metabolic rate. So, when you do workouts that require more oxygen not just during the training but also after you've stepped out of the gym, you'll burn more energy—in other words, calories. No prizes for guessing which type of workout accomplishes this—weight training!
- **Strength:** This one's a no-brainer! The more you lift, the stronger you get.

Myth #2: Women should not lift weights!

The one question I keep getting asked again and again is this—should women lift weights? Women, sisters, mothers—lend me your ear!

The answer is a resounding YES!

Sadly, everyone, from gym trainers to Instagram Influencers, wants to tell you that women should only do cardio. If you're going to lift weights, make sure you stay away from the heavy weights—those tiny pink dumbbells are good enough for you.

The reason: lifting weights will make you 'bulky'. Your aim should be to look 'toned'.

The myth is that if you lift heavy weights, then you'll lose your femininity and start to look muscular like a bodybuilder.

Nothing could be further from the truth.

Here's why women should lift weights:

You will look lean, not bulky: Testosterone is the principal hormone that drives muscle growth in men. What testosterone is to men, estrogen is to women. Women may have a smaller muscle mass to begin with but don't let that fool you. When women start lifting weights, they have the potential to make gains in muscle mass similar to men.

Read that again: Women have the same muscular potential as men.

This does not mean that women will walk around looking like She-Hulk if they start lifting. It simply means that with a proper workout and nutritious diet, women can expect to make the same percentage gain in muscles as men. And when your muscles are properly developed, that spells strength, energy and vitality.

Besides, so many women pick up their kids or grocery bags and carry them around which definitely weigh much more than those tiny pink dumbbells. How often do you see them with unnaturally bulging biceps?

Lifting weights help women stay fit: When you lift light weights (usually accompanied by high reps), this may improve your muscle endurance, but it won't do much for giving you that svelte yet strong look that you crave. For that, you need to build lean muscle which is the result of proper training and nutrition. And the key to building muscles—lifting heavy!

Case Study: My Own Grandmother

As you know, I grew up in a village. We didn't have the fancy gadgets and household items that so many people living in cities today take for granted. There were no mixer-grinders or washing machines—we did everything with our bare hands and that included my old grandmother. Besides, my grandmother was old-school, and she did what women in India had been doing since time immemorial—she lifted heavy.

Let me explain.

Every week or 10 days, my grandmother would haul a heavy stone mortar and pestle into our small courtyard. She would then adjust her pallu over her head, cover her mouth and nose, and start pounding spices and grains to make masalas and flour for the entire household.

With one hand, she would add the red chilis or wheat grains into the mortar and with the other hand—with just one hand, mind you—she would rhythmically throw the pestle into the mortar with great force and crush the whole ingredients into fine powder.

She'd do this first with her right arm and, as that side grew tired, she'd switch hands and use her left arm to do the same thing. Over and over, all the while standing ramrod straight.

And that wasn't all—my grandmother was also in charge of all the chutneys and wet masalas. For that, she used a flat stone platform and a rounded stone crusher which she would lift all by herself, use without any assistance and then, wash and keep away for another day.

My grandmother lived to a ripe old age and almost till the end of her life, she continued with this practice. She was fit, and healthy—free of aches and pains. The reason: she lifted weights.

I encourage all the women reading this to do the same.

Lifting Weights and Old Age

If all this still hasn't convinced you that you need to lift weights, this will definitely be an eye-opener:

Far too many women (and men) experience osteoporosis and sarcopenia as they age.

Osteoporosis is a condition where the bones weaken over time and become brittle. This increases the risks of fractures and in some extreme cases, even simple things like sneezing too hard can result in bone breakage.

Sarcopenia is age-related muscle atrophy or muscle loss. It's a degenerative condition where the muscles grow weaker and smaller. In fact, without the right intervention, you could end up losing 10 per cent of your muscle mass every decade starting from the age of 40.

The way to safeguard your health in your silver years is to lift weights when you're younger. Lifting weights along with a proper diet can go a long way in keeping you healthy and preventing these age-related conditions which can severely hamper your quality of life.

Don't worry if you've never lifted weights—start now. It's never too late to reap the benefits of weight training.

A FINAL WORD: PRINCIPLES OF EXERCISE

So, now you've decided to start with resistance training.

But how do you keep track of your workout progress and ensure that your workout routine continues to work for you?

All successful programs follow certain basic principles; knowing them will help you understand why you exercise and how to judge whether or not you are making progress. That way, you'll not only get the most out of your exercise program but also stay focused on your journey.

These principles are the roadmap on your fitness journey which you need to keep consulting to check if you're headed in the right direction.

Principle #1: Specific Exercises for Specific Fitness Goals

To get the physique goals you want, you need to perform those specific exercises which will help you achieve that goal.

If your goal is to run the marathon, you need to train specifically for it. You need to train in a way that places your body in the same situations or stresses that you would endure while running a marathon.

Let's take another fitness goal: fat loss. Your aim is to lose excess fat without losing your lean muscle mass. Therefore, you need to choose an exercise program that is specifically designed to optimize fat loss.

Principle #2: Find an Exercise/Workout Program That Works for YOU

We've seen this principle in the previous chapter as well.

Let's say you and your friend follow the same diet. Will both of you see the same results? Not likely. Our bodies are unique, and each person responds differently to a diet. That's why it's essential to design a customized nutrition plan that is suitable for your specific situation.

In the same way, you can't follow the same exercise plan as your friend and expect to see the same results. Even if both of you lift the same amount of weight, do the same exercises for the exact number of sets and reps and even train for the same number of days each week, the results both of you will see are likely to be quite different.

It's quite common these days to read articles about your favourite superhero's exercise program. Fanboys (and fangirls) blindly start following this routine and think that they too will look like their idols. Don't be disappointed when you look nothing like Captain America or Black Widow even after exercising within an inch of your life and sanity.

Find what works for you. And stick to it!

Principle #3: Keep Progressing in Your Workouts

As in life, you will grow and keep making progress with your chosen exercise program only when you continually push the limits of what you can do.

To make progress with your workouts, you need to increase your body's workload gradually. But it's not as simple as 'lift heavier weights each time you go to the gym'.

You also have to ensure that you don't burn out or fatigue. Following an exercise program that's too taxing on your body would be counterproductive—instead of making progress, you'll open up your body to the risk of injuries that could keep you out of action.

For sufficient progressive overload, you need to tweak three factors of your exercise program—Frequency, Intensity and Time.

Principle #4: Find a Workout Routine That's Flexible

Follow an exercise program that leaves room for your life. Stuff comes up all the time, and sometimes, we need to take a break or skip a workout or two. And that's perfectly alright, so long as you don't let a missed workout turn into a missed month. If your program is too rigid or stringent, chances are you'll keep falling off the wagon, hate yourself and never make it back to your routine. That's why it's important to be realistic about your schedule and how you can fit exercise into it. Follow an exercise program that's flexible enough, and you're more likely to stay on course.

Another important aspect of keeping things flexible is to include variety in your workouts. As we'll see later on in this chapter, different goals need different types of exercise routines. But that doesn't mean you have to stick to just one kind of exercise routine. Feel free to mix things up, and you'll find that you end up enjoying your workout sessions even more, besides seeing better results.

Principle #5: Long Term Goals Are Better

You've made a new year's resolution to get fit, and you know exactly how you will achieve that goal: by working out twice a day. You think you're doing everything right—and for a while, things look sunny.

But soon, you find that you're constantly feeling tired and run down—not at all how you expected to feel.

You, my friend, are fatigued and possibly overtrained.

Overtraining usually occurs when you're not giving your body enough time to recover from your workouts. Without adequate rest and recovery, you become prone to fatigue and injury. Your progress slows down or can even stall. But don't think that only people who lift weights at the gym can overtrain.

Marathoners who run a half marathon each weekend, cyclists who push themselves farther each day, yoga enthusiasts who practise for hours—all of you could end up overtraining.

How can you avoid such a reversal of fate and fitness?

- By being consistent.
- By listening to your body and giving it adequate rest.
- By realizing that your health journey is a marathon and not a sprint.
- By taking things one day at a time and not overdoing them.

Build your workout routine slowly and gradually. Stick to it for long enough. Results will follow.

It's as simple as that.

To sum up, a successful exercise plan has the following features:

- Goal specific
- Fits your lifestyle
- Enjoyable
- Allows you to progress
- Leaves room for life
- Leaves time for recovery
- Should be challenging (but achievable)

The next chapter is what you've been waiting for. We will bring these principles into practice and create your own personalized workout plan—one that will get you started on your fitness journey and help you get into good shape.

References

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PREPARE YOUR OWN WORKOUT PLAN

I'm extremely excited about this chapter. If I had known these concepts and started lifting weights properly when I was a young lad, I would have seen great results so much sooner. But it doesn't matter how old you are or how much workout experience you have—the important thing is that you have decided to start lifting weights and I'm here to help you get started.

I know that preparing one's own workout plan isn't everyone's cup of tea, which is why I've included a couple of sample workout plans in this chapter. If you wish, you can just pick one of them. However, I would urge you to take some time to read through this chapter and create your own workout chart. This won't just help you understand why certain exercises are included in your plan but will also help you create new workout plans of your own as you progress.

Let's start with some key basics.

Components of an effective workout plan

All good workout plans have a few things in common. If you want to make the most of your workout, make sure you incorporate these without fail:

1. Warm-up: Don't start your workout without spending at least five to 10 minutes getting warmed up. Dynamic stretching, mobility exercises, performing exercises with light weights or doing the bodyweight versions of each exercise before picking up weights are just some ways to warm up. Start light so that you can go heavy.
2. Conditioning: This is the main workout. Spend anywhere from 30 to 60 minutes doing your exercises. Pay close attention to form and technique if you want to avoid injury and derive the maximum benefit from the workout.
3. Hydration: You don't need any fancy drinks, plain ol' water is good enough. Keep taking sips throughout your workout.
4. Cool-down: This is an often-neglected part of your workout. After finishing your training, spend five to 10 minutes doing some static stretches. Stretching allows your muscles to revert to their original state.

Workout tips for beginners

- . Pay attention to your form and technique: Whether you're a beginner or more advanced, the best way to progress and to prevent injuries is to pay close attention to proper exercise form. It's a good practice to watch workout videos from reliable sources to understand how an exercise should be executed. On the Fittr app, we have more than 300 exercise videos with complete and step-by-step instructions, so you might want to check them out before hitting the gym. Also, while you're at the gym, just ask a trainer or member to check if you're doing an exercise correctly. Don't feel shy, you'll be surprised at how willing most people are to help out someone with their workouts.
- . Stick to the same workout: You need to keep doing the same exercises over and over again till they become second nature. Don't change your workouts too often, stick with them for longer durations of time. This will help you build strength and also get better at your exercises, so that you can start lifting heavier weights over time.
- . Slow and steady: Don't be in a rush to start lifting like that hulk who's been working out for years. He didn't (hopefully) jump into the thick of things and just start lifting heavy. Start with lighter weights and simpler variations of exercises. Then, slowly progress to more complex variations.
- . Machine-based exercises: If you decide to join a gym, don't feel daunted by all those shiny machines. They're excellent for beginners because they help you with your posture and you can easily and quickly adjust the weights if you think they're too heavy. Here's a tip: most of these machines have a sticker on the side which tells you which exercise it should be used for, which muscles you'll be targeting and they even have step-by-step instructions and diagrams so that you know how to use them. However, if you don't have access to machines, go ahead and use barbells and dumbbells.
- . Lift neither too heavy nor too light: Here's a simple way to decide: if you lift a certain weight (say 12.5 kg) and you can easily do 15 to 18 reps, then the workout is probably too easy. Lift a slightly heavier weight (say 15 kg) and aim for eight to 12 reps, such that the last couple of reps should feel slightly difficult to complete. Of course, don't lift too heavy and always, always pay attention to your form and technique. If you're not able to perform an exercise properly because the weight's too heavy, immediately switch to a lighter weight.
- . Don't skip any exercises: If you cannot perform an exercise, find the alternative version but don't remove the exercise from your routine. As you move through the weeks, keep practising the tougher ones till you 'get it'.
- . Don't rush through your workout: You're not running a sprint, you won't get any bonus points for finishing your workout before anyone else. Make sure you're taking adequate rests between each set. Do not rush through the session. And as always, focus on form and technique.
- . Train at least two muscle groups per day.

Now, it's time to get cracking and create your own workout plan.

Foundations of a workout plan

When you sit down to create your workout plan, there are five factors you need to take into account first:

1. What is your goal?
2. Which types of exercises/workouts do you need to do to achieve this goal?
3. How many days per week will you work out?
4. Have you given yourself enough time to recover?
5. How do you plan to progress? (i.e. continue to lift heavier and grow stronger over time)

Like we did for the nutrition plan, I have also created a two-page workout plan template for you to use. As we learn some important concepts related to workouts and exercise, we will simultaneously fill up this template. This is the best way to understand the concepts and apply them in real-time. And, by the end of this section, you will have your very own workout plan!

Part A: Personal details

WORKOUT PLAN	
PART A – BASIC INFORMATION	
Personal Details	
Name:	
Age:	
Height:	
Weight:	
Key Details	
Step 1 — My Fitness Goal	
Write your goal here:	
Step 2 — My Key Numbers	
Body Fat %:	
Body Measurements (in inches):	
a) Chest	
b) Bicep	
c) Thigh	
d) Waist	
e) Hips	

In this section, fill in your details.

Name, age, weight, height, body fat percentage/body circumferences measurements (chest, bicep, thigh, waist, belly and hips)

What is your goal?

How many days per week can you work out?

What are your rest days?

Part B: Workout creator

PART B – YOUR PERSONAL WORKOUT PLAN			
My Workout Split: _____			
My Workout Days : _____			
My Rest Days : _____			
Exercise	Sets	Reps	Rest Between Sets
DAY 1			
DAY 2			
DAY 3			

In this section, we will learn and apply the key concepts involved in creating a workout plan.

Workout protocols/Splits

Workout splits is a method of organizing your exercises for the week into a sustainable and achievable daily plan. As we saw earlier, it's important to allow your body to recover from the previous workout. Using a workout split is a great way to ensure that you hit your weekly workout targets and also avoid fatigue.

Here are four popular methods to program your workouts:

Method #1: Push/Pull/Legs

This is a great workout split to follow. For your Push days, you will train the pushing muscles of your upper body—chest, shoulders and triceps. This means you will need to perform exercises that require you to push weight away from your body. Eg. Chest Press, Overhead Shoulder Press, Skull Crushers for triceps, etc.

Similarly, on your Pull days, you will perform exercises for the upper body which employ the pulling muscles (viz. Back and Biceps) and where you pull weights towards your body.

Your weekly plan might look something like this:

Alternative 1

Monday: Push workout
Tuesday: Pull workout
Wednesday: Legs workout
Thursday: Push workout
Friday: Pull workout
Saturday: Legs workout

Alternative 2

Monday: Legs workout
Tuesday: Pull workout
Wednesday: Push workout
Thursday: Pull workout
Friday: Legs workout
Saturday: Push workout

Alternative 3

Monday: Pull workout
Wednesday: Legs workout
Friday: Push workout
Tuesday, Thursday, Saturday: You can do cardio, yoga or any other workout, or you can use them as rest days.
Sunday: Either rest day or an extra workout, depending on your preference.

Method #2: Upper/Lower

This split works out the same body parts/muscles more frequently. It also allows you to try different variations of the same exercise or different exercises that employ the same muscles. If you like variety in your workouts, then this split might be for you.

Alternative 1

Monday: Upper body
Tuesday: Lower body
Wednesday: Upper body
Thursday: Lower body
Friday: Upper body
Saturday: Lower body

Alternative 2

Monday: Upper body (Push emphasis)
Tuesday: Lower body (Free weights)
Wednesday: Upper body (Pull emphasis)
Thursday: Lower body (Machine weights)
Friday: Upper body (Push and pull)
Saturday: Lower body (Bodyweight exercises/Resistance bands)

Method #3: Full Body Daily

If this business of doing workout splits seems too complicated, just do a full body workout anywhere from three to six days a week.

Alternative 1

Monday: Full body
Wednesday: Full body
Friday: Full body
(Do 2-3 exercises for each muscle group.)

Alternative 2

Monday: Full body (Push emphasis)
Tuesday: Full body (Pull emphasis)
Wednesday: Rest day
Thursday: Full body (Push emphasis)
Friday: Full body (Pull emphasis)
Saturday: Active rest day (This means that you don't work out on this day but rather, remain active by doing some walking, swimming or running.)

Sunday: Rest day

Method #4: Back and biceps/Chest and triceps/Legs and shoulders

This is one of the most common workout splits used the world over. It's easy to remember, easy to implement and helps you work out every single muscle group during the week, while also giving you enough time to recover. Also a good split for lovers of variety in exercises.

Alternative 1

Monday: Back and biceps

Tuesday: Chest and triceps

Wednesday: Legs and shoulders

Thursday: Back and biceps

Friday: Chest and triceps

Saturday: Legs and shoulders

Sunday: Rest day/Active rest day

Alternative 2

Monday: Back (push emphasis) and biceps (free weights)

Tuesday: Chest and triceps (both machine weights)

Wednesday: Legs and shoulders (both free weights)

Thursday: Back (pull emphasis) and biceps (machine weights)

Friday: Chest and triceps (both free weights)

Saturday: Legs and shoulders (both machine weights)

Sunday: Rest day/Active rest day

In the Template: Decide the number of days you will be working out. To begin with, pick one workout split. In the next section, we will select the exercises that go with each split.

Exercise selection, volume, intensity and frequency

Once we've decided the workout protocol and the number of days we will be working out, the next step is to select exercises. We aim to select exercises that will work out every major muscle group.

In the Template: Add the exercise names to your sheet. For example, you've decided that you'll be working out three days per week (Mon/Wed/Fri), then your workout could look something like this:

SAMPLE WORKOUT PLAN - BEGINNER			
My Workout Split:		Chest & Back/Legs & Abs/Arms & Shoulders	
My Workout Days:		Monday, Wednesday, Friday	
My Rest Days :		Tuesday, Thursday, Saturday, Sunday	
Exercise	Sets	Reps	Rest Between Sets
Day 1: Chest & Back			
Dumbbell Bench Press	4	12-15	45 seconds
Lat Pull Down	4	12-15	45 seconds
Incline Barbell Bench Press	4	8-12	45 seconds
Seated Row	4	8-12	45 seconds
Bent Over Row	4	12-15	45 seconds
Day 2: Legs & Abs			
Barbell Full Squats	4	8-12	45 seconds
Leg Curl	4	12-15	45 seconds
Leg Extention	4	15	45 seconds
Calf Raises	4	20	45 seconds
Plank	4	Hold as long as possible	45 seconds
Day 3: Arms & Shoulders			
Dumbbell Shoulder Press	4	10-12	45 seconds
Dumbbell Alternating Lateral Raises	4	12-15	45 seconds
Barbell Shrugs	4	8-12	45 seconds
Cable Biceps Curls	4	12-15	45 seconds
Dumbbell Lying Tricep Extension	4	12-15	45 seconds

The next step is to decide how much of each exercise to do. It's not enough to be regular at the gym. In addition to that, you need to progress and scale up your workouts over time. Scaling up means changing specific parameters of your workout over time so that it continues to challenge you. This will provide your muscles with the stimulus they need to grow. This discussion is a

little technical, but as you'll be lifting weights soon, you need to understand these key concepts.

A successful training plan optimizes training volume, intensity and frequency in such a way that it allows the individual to keep getting better week after week and also allows sufficient time for recovery.

. **Training volume:** Volume is the total amount of work done. It is generally calculated for each muscle group. There are different ways of calculating the workout volume:

- Volume load = Sets $\times$ Reps $\times$ Load
- Repetition volume = Sets $\times$ Reps
- Sets volume = Number of sets

Reps: A rep is completed when you perform an exercise through the full range of motion (ROM), for one time. For example, if you're doing bicep curls, then a full ROM would mean lifting the barbell/dumbbell, contracting your bicep muscle as you lift the weight and then bringing the weights back to the starting position. This completes 1 rep.

Sets: A collection of reps done one after the other. For example, if you did 8 reps of the bicep curl before stopping, then those 8 reps constitute 1 set.

Load: The amount of weight lifted while performing the exercise.

. **Training intensity:** 'My workouts are really intense,' a guy at my gym once told me. What he meant was that he made a lot of grunting noises while lifting weights and always made sure to throw the weights down once he was done.

That's not what training intensity means. There are several ways to measure training intensity, such as Percentage of 1 Rep Max, Rating of Perceived Exertion (RPE), Reps in Reserve (RIR). All these measures try to gauge the intensity of your effort as you perform the exercise as they consider your lifting load and judge how close to your maximal effort that load is.

. **Training frequency:** This one is pretty easy to understand. This measures how frequently you train a particular body part/muscle group in a week.

Recommendations based on research suggest that training volume for beginners should be six to 12 sets per muscle group or movement, per week. Intermediate lifters should aim for 13–20 sets per muscle group or movement, per week. Intensity should be in the six to 15 rep range for muscle building and in the one to six rep range for strength. The suggested frequency is training of each muscle group or movement pattern at least twice a week. Please note that these are approximate values. Choose your own variables, as per your abilities.

In the Template: In front of each exercise, add the target sets and reps. Leave the weight field blank, you will fill this up at the gym.

How much weight should I lift?

One rule of thumb is to pick a weight that leaves one rep short of failing your set. If you lift too light, you'll not be making adequate effort, and you won't be providing sufficient stimulus for the muscle to grow. If you lift too heavy, you'll not be able to complete the required workout volume. Besides, your form and technique might get compromised, which increases the risk of injury. Remember, NO ego lifting!

And that's it! Your workout plan is ready. Carry this sheet with you to the gym when you work out. Make a note of the weight lifted during each set. This will help you track your workouts over time.

Some additional things to consider

You need to give your body sufficient time to recover from the stress of a workout. This will prevent injuries and fatigue and help you come back stronger each time you lift weights. Keep these points in mind:

- . **Rest intervals during the workout:** Generally, 60 to 90 seconds of rest between sets is sufficient. Research shows that if you rest for at least 2 minutes between sets, it can improve your ability to maintain greater volume load.

Here's a general rule of thumb: rest until you've managed to catch your breath or feel ready to hit the next set as long as it's more than 60 seconds.

- . **Rest days:** Allow sufficient rest days between two workouts. If you're just starting, you could work out only three days a week, every alternate day (Mon/Wed/Fri or Tues/Thurs/Sat). The other days can be your rest days when you don't lift weights. If you wish, you could utilise your days off from the gym to do cardio, yoga or some other form of exercise that isn't too heavy.

As you grow stronger, you can start working out for more days. I work out six days a week, but I've had many years of practice. If you're just starting out, allow your body to recover from the previous workout before doing the same workout again.

Let's take an example: if you do a chest workout on a Monday, then don't do another chest workout the very next day. Allow your chest muscles to recover from the previous workout. Do some other workout on Tuesday (say, legs) and do your second chest workout of that week on Wednesday.

- . **Sleep:** Aim for seven to eight hours of sound sleep every night. Exercise is a form of stress for your body, and it needs to recharge itself every night. Lack of sleep or poor sleep hygiene can hamper your workouts and slow down your progress.

Listen to your body. If you're feeling tired, it's okay to skip a workout or perform a lighter-than-usual workout.

Bodyweight workouts or Workout from Home

2020 was an interesting year, wasn't it? Post-lockdown, we're still hearing and speaking words such as 'the new normal'. The pandemic had two major fallouts as far as people's attitudes towards health and fitness are concerned. First, people finally realized the importance of health and fitness, and second, there was an increased interest in finding ways to work out at home.

Gyms were closed, people were cooped up at home, and many of us needed an outlet for all our pent-up energy. As work from home became a never-ending reality (or nightmare), people needed something to do. Enter bodyweight workouts. #WorkoutFromHome was a hot and trending topic for months. Left with no choice, people had started to scan the Internet and fitness apps for videos on how to perform the perfect push-up.

And guess what? A lot of people found out that home workouts work! So, it would be remiss of me not to include bodyweight workouts.

How do bodyweight exercises work? Bodyweight exercises such as push-ups and pull-ups use your body's weight to create resistance for your muscles. When you perform a push-up, you're working against gravity which creates tension in your arm muscles and gives you a good workout.

The beauty of bodyweight exercises is that you can do them anytime, anywhere and you don't need any additional equipment. What's more, you can even use household items such as backpacks and water bottles to add some more weight to your exercises. Almost any exercise that you can perform at the gym with weights has a corresponding bodyweight exercise.

How to plan home workouts

- . Assess your current fitness level—beginner, intermediate or advanced—and accordingly incorporate exercises. If you're new to exercise, start with a simple or basic version and add variations as you get better and stronger.
- . Include exercises for all muscle groups.
- . Schedule the exercises in descending order of their difficulty level. Complete the more challenging exercises first.
- . There are several ways to add progression/periodization.
 - Do more sets or reps of the same exercise.
 - Add external weights such as backpacks filled with books or bottles, buckets filled with water, etc.
 - Add variations (progress from basic push-ups to single-handed push-ups). The chart below will serve as a guide.

To get you started, here is a sample home workout plan using minimal to no equipment.

Sample Home Workout Plan (Beginners)

SAMPLE WORKOUT PLAN - BEGINNER			
My Workout Split:	Upper / Lower		
My Workout Days:	Monday to Saturday		
My Rest Days :	Sunday		
Exercise	Sets	Reps	Rest Between Sets
Day 1, 3 & 5: Upper Body			
Push-up	4	15	45 seconds
Incline Push-up	4	10- 12	45 seconds
Dumbbell Bent Over Row	4	15-20	45 seconds
Dumbbell Shoulder Press	4	12-15	45 seconds
One-Arm Dumbbell Row	4	12-15	45 seconds
Dumbbell Raises	4	12-15	45 seconds
Dumbbell Bicep Curls	4	15-20	45 seconds
Dumbbell Lying Tricep Extension	4	15-20	45 seconds
Day 2, 4 & 6: Lower Body			
Split Squats	4	8-10	45 seconds
Lunges	4	10- 12	45 seconds
One Legged Deadlifts	4	10- 12	45 seconds
Sumo Goblet Squats	4	12-15	45 seconds
Glute Bridges	4	20-30	45 seconds
Single Leg Calf Raises	4	12-15	45 seconds
Plank	4	Hold as long as possible	45 seconds
Ab Crunches	4	20-30	45 seconds

If you don't have access to a gym or even the most basic equipment, that should not stop you from getting a good workout. I guess even the pandemic had a silver lining.

A Final Word

You only have this one body. Take care of it as if your life depends on it—because it does! Take it one step at a time, read and re-read this book and then, go out and apply these lessons.

You're now officially part of the Fittr revolution that has impacted thousands of lives across the globe. Chart your own success journey. Create your own transformation story. Live your best life.

Your fitness journey begins now!

SUPPLEMENTS

Tall Claims, Myths...Lies?

A few weeks ago, a supplement company sent me some product samples. I'd never heard of this company and I'd certainly not solicited their products, but when you're the founder of a fitness company, people send you their stuff hoping you'll promote them. It was a standard set of your typical supplements, nothing out of the ordinary. But as I read the labels, something caught my eye. These supplements were not any run-of-the-mill supplements—they were 'especially made' for the Indian body!

My entire nutrition and fitness education flashed before my eyes. Had I missed something in the thousands of pages that I had read? Was the Indian body different from the bodies that are found in other countries? What was so unique about our anatomies that the supplements that are manufactured in the USA or UK wouldn't work for us? Needless to say, this was just a marketing gimmick to make these very ordinary run-of-the-mill supplements stand out from the competition.

This is just a very small example of the kind of lies that some not-so-ethical supplement companies have been shoving down our collective throats. And compared to some of the claims that I've seen over the years, this one wasn't even all that nefarious.

Morning Walks and Health Tonics

A lot of people in India love to go for morning walks in community parks. It's a great time for people to meet their gang and catch up on the latest gossip. Moreover, green spaces have become so rare that it's become the only time to get some clean air into your lungs.

Across the country, you'll see a familiar sight at these parks: a small table with thermos flasks or steel drums filled with a variety of juices. They have everything from karela (bitter gourd) to aloe vera, wheat grass to 'immunity-boosting' ginger, neem and amla. After finishing their walks, people throng these makeshift kiosks to down a shot or two of these healthy drinks.

Nothing wrong with that. But have you ever wondered if they really help? Bitter gourd is supposed to help with diabetes. Truth or just another fitness myth?

I have spoken with a number of doctors, of both allopathy and Ayurveda, and they all had one thing to say: these 'health potions' are useless!

Should you stop drinking them? No, not really. If you like the taste, go right ahead. It certainly beats drinking fizzy sugar water. But if you expect them to change your health and make you fit, think again. For that, you still need a proper diet and exercise.

Tall Claims

Go to any online health store and type in the word ‘supplements’. Chances are you’ll run into product suggestions like garcinia cambogia, green tea, green coffee, fat burners and mass gainers. What are these wondrous products that companies want you to purchase? They promise to burn all your fat, help skinny people gain weight or give you so much energy that you’ll put the Duracell bunny to shame.

Enrol with a gym trainer and chances are he will suggest that you take some fat burners to ‘speed up the fat loss’. And let’s not forget our friendly neighbourhood supplement company representatives who can’t stop pushing their meal replacement potions and powders which ‘provide you with all the nutrients your body needs, so why eat food which is full of calories?’

Come to think of it, even the chocolate-flavoured malt drink that you’ve been adding to your milk since childhood (you know which one I mean) is a supplement that is supposedly a must-have for all growing boys and girls. These days, there are even supplements and multivitamins that are exclusively made for women.

All these products are pushed onto an unsuspecting public through advertising, product placements and a constant bombardment in media about their benefits. The agenda is simple—to give you an intense feeling of FOMO and to make you feel guilty or fearful for your health if you don’t take them. But what is the truth about supplements?

What Are Supplements?

Supplements, as the name suggests, are supposed to *supplement* your diet and either help you consume the required amounts of macronutrients or micronutrients so that you stay in good health, or provide you with bigger doses of certain nutrients to fuel athletic performance. They are *not*—and I cannot stress this enough—a *replacement* for food.

Just because you have an insurance policy, is it okay for you to start driving in the middle of the road in the opposite direction? Or start smoking like a chimney? No, of course not. In the same way, just because supplements exist does not mean you can neglect your main source of nutrition—real, whole food.

Who Needs Supplements?

Are you ready for the answer? Here it is—no one!

You read that right: nobody *needs* to take supplements. The word itself implies that it is not a need but something you can take, if you think it can provide you with some additional benefits. You still need to follow a healthy diet. Supplements will help you stay in good health, provided you have all the other things in place.

For example, as we will see later in this chapter, whey protein is an easy way to complete your protein macros. But does that mean you *need* to consume it? No. You can—and should—try to complete your protein macros from real food sources. But if you're finding it difficult to get sufficient protein from your diet, then whey can help you reach your daily protein target.

It is important to have the right attitude towards supplements—they are not a panacea for your poor health choices, so don't rely on them too much. They are an aid, not a crutch.

How Do Supplements Help?

Not all supplements are useless. The right ones can help in a number of ways.

- They deliver a concentrated form of nutrients (such as whey protein) and help you complete your macros.
- If allergies prevent you from consuming certain foods, then supplements can provide you with the required nutrients and mitigate deficiencies.
- People on low calorie diets or who have poor digestion can benefit from supplements as these products often fit in your macros and some even contain digestive enzymes.
- Athletes who need high quality nutrition in the correct doses and for whom timing matters can use supplements to augment their performance.
- The biggest benefit: convenience!

Supplements Are neither Evil nor Magic

On the other end of the to-supplement-or-not-to-supplement spectrum lie the fearmongers and naysayers.

Their stories range from ‘supplements are steroids’ to ‘supplements will kill you’. Frankly, I don’t know which stories are more hilarious, claims that supplements are dangerous or the tall tales of how supplements are magic. As usual, the truth about supplements can only be answered when one understands the science.

Understanding Supplements

In this chapter, my aim is to demystify some of the popular (and a few not so well-known) supplements that most of you will encounter on your fitness journeys. I have divided supplements into the following categories:

1. Helpful supplements: Supplements that are backed by science and solid research. They have been shown to be helpful, relevant and are definitely worth considering.
2. Natural/Ayurvedic supplements: Those which are backed by science and that actually work.
3. Hyped/No-good supplements: The con artists of the supplement industry. Steer clear of these top offenders.

One of the best resources for research on supplements is examine.com, which was started by Kamal Patel. This website is my go-to resource for the latest scientific research, papers and findings on supplements and should be on your reading list too if you want to stay updated on what's new in nutrition science. Most of the research that I've used in this chapter can be found on this website.

Let's get started.

1. Helpful supplements

i. Whey Protein

Even if you don't have the slightest knowledge of supplements, I'm sure you've already heard of whey protein. Walk into any medical store or supplement store and you'll encounter these ubiquitous tins and packs of whey. Gym culture would be incomplete without some dude or dudette casually sipping a frothy liquid from a plastic shaker. These days, you even get USB-powered blenders that you can use for whipping up a protein shake on the go.

What it is: Whey is derived from milk protein and is a by-product of the cheese-making process. In fact, it is even created when you make paneer or curd at home. Have you ever noticed that watery layer on top of your dahi or paneer? That's whey.

On a large scale, whey is created by adding special enzymes to milk, which separates curd and whey. This whey is then filtered or pasteurized to remove impurities and spray dried into a powder form. After adding flavours, digestive enzymes and sweeteners, this powder reaches the market as 'whey protein'.

There are three popular types of whey protein that are commonly sold: whey protein concentrate (WPC), whey protein isolate (WPI), and whey protein hydrolysate (WPH). Of these, WPC is the least refined, and therefore, the cheapest form of whey available in the market. However, inexpensive does not mean ineffective. WPC delivers a high amount of protein per scoop (upto 80 per cent of product weight) but it also tends to contain slightly higher amounts of carbohydrates and fats.

WPI is essentially WPC which has been further filtered to remove lactose and extra carbohydrates and fats. Hence, it tends to be a favourite for calorie conscious individuals and those on a low carb diet. Its protein availability is around 90 per cent of product weight. In addition, removal of lactose makes it more easily digestible. But it also tends to cost more than whey concentrate.

WPH contains additional enzymes which break up the protein into easily digestible peptides. This variant is, by far, the most expensive one.

So which one should you buy? Whey concentrate is good enough. If you're lactose intolerant, then whey isolate may be a better fit but it will also make a bigger dent in your wallet.

In recent years, the Indian market has also opened up to vegan varieties of protein powders such as pea protein, rice protein and hemp protein. There are also organic whey powders, grass-fed whey protein and all sorts of highfalutin varieties. I haven't tried them and I'm not likely to do so. Feel free to buy them if you can afford them (they're significantly more expensive than the ordinary whey proteins) but know this: while they may or may not be better for the planet, they'll yield only as much protein as the regular varieties.

How it works and benefits: The benefits of protein for the body have already been discussed. Whey protein can make it easier for some people to complete their daily protein macros, especially if they're on a low caloric diet or simply don't have the appetite for more food.

Dosage, frequency and timing: A typical scoop of whey is about 30 grams and yields anywhere from 20–25 grams of protein. It's a myth that you should not have more than one to two scoops of whey per day, you can have more scoops if you want to, it is safe! It is also a myth that if you don't consume whey protein within 20 minutes of a workout, your muscles won't grow (the myth of the 'anabolic window'). You can have whey protein before a workout and any time after a workout. You can even consume whey protein on the days when you don't work out, if it helps you complete your protein macros.

Can you take it every day? Yes, you can.

Is it safe for long term use? Yes. However, as already discussed, try and get your protein from whole foods as much as possible.

Possible side-effects: Some people might experience flatulence or find it difficult to digest whey protein. Mix the protein powder in about 500 ml of water and stay hydrated throughout the day—that should take care of the gassiness! Also, choose a protein powder that has added digestive enzymes. If you are lactose intolerant, then you can even switch to pea protein or soy protein (provided you're not allergic to soy).

ii. Multivitamins

When we were kids, a lot of us must have read in school textbooks about diseases such as scurvy or rickets which are caused due to vitamin and mineral deficiencies. Most of us have never encountered these once-common illnesses because the quality of our foods has vastly improved over the ages. Plus, many foods are now fortified with micronutrients, which makes it difficult

for someone to experience a deficiency. So, does it still make sense to take a multivitamin?

What it is: A pill, tablet, capsule or even a syrup or tonic that is a preparation containing a variety of vitamins, minerals and other vitally important micro elements that are needed for maintaining optimal health. The idea behind taking a multivitamin is that one does not experience any deficiencies.

How it works and benefits: While a healthy and balanced diet is still our best bet against deficiencies, we may sometimes be unable to get the full spectrum of vitamins and minerals solely from our food. A multivitamin is a good precaution to take to ensure this risk is mitigated.

Dosage, frequency and timing: One multivitamin tablet per day is usually all you need. Take it at breakfast. Choose a good quality multivitamin that contains close to 100 per cent of the RDA values of all vitamins and minerals.

Can you take it every day? Yes, you can.

Is it safe for long term use? Yes, it is.

Possible side-effects: None.

iii. Fish oil (EPA and DHA)

This is another well-known supplement which doesn't need any introductions. As children, many of us must have been force-fed cod liver oil capsules by our concerned parents which we hated because they gave us weird-tasting burps. Turns out, our parents were doing the right thing—for once!

What it is: Fish oil is a rich source of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), two of the most vital essential fatty acids that our bodies need.

How it works and benefits: Just some of their benefits include reduced symptoms of depression, lowered risk of heart disease, reduced blood pressure, and even decreased waist circumference. While more research is needed, it may also improve body composition (more muscle and less fat) and performance in the gym.

Dosage, frequency and timing: 1–3 g of EPA and DHA are considered adequate and there is no evidence that taking more than that dose provides any additional benefits. If seafood is a part of your diet, you can skip the fish oil capsules or limit their intake to days when you don't eat fish.

Can you take it every day? Yes, you can.

Is it safe for long term use? Yes, it is.

Possible side-effects: None. Some people complain of fishy burps. Drink plenty of water and the problem should go away. If not, change the brand.

iv. Zinc

What it is: Zinc is an essential mineral which plays a number of key roles. It's mostly found in

meat, eggs and legumes, so unless you regularly eat these foods, you might face the risk of zinc deficiency.

How it works and benefits: Zinc has been proven to be vital for the activity of more than 300 enzymes. These help metabolize macronutrients and also aid cell replication which is a key biochemical function that corresponds to recovery and growth. Zinc has also been shown to have positive effects on the anabolic hormone profile, particularly testosterone. It increases free serum testosterone levels which is particularly important in older men as their testosterone levels start to decline with age.

Another important benefit of zinc is that over time, it boosts the immune system and can reduce the frequency of illnesses.

Dosage, frequency and timing: There are two standard doses for zinc: 5–10 mg per day (low dose) and 25–45 mg (high dose). You don't really need to take the high dosage if you don't have a zinc deficiency. A low dose is more than enough as a preventative measure.

Can you take it every day? Yes, you can.

Is it safe for long term use? Yes, it is.

Possible side-effects: None. Just make sure you stay within the prescribed range.

v. Caffeine

One of the things that gym bunnies love doing is sipping a pre-workout before they head off to the gym. Combined with some headbanging music, it makes them feel really bad-ass! I don't know about the music bit but I can attest to the fact that when the pre-workout kicks in, you feel amazing!

Those among us who don't have money to burn can simply down a cup of black coffee before our workout and reap some of the same benefits at a much lower price point. The reason: caffeine!

What it is: A very popular supplement and ingredient in pre-workouts.

How it works and benefits: Caffeine works in two ways: it suppresses tiredness and it also acts as a stimulant and enhances performance. Research studies have found that when caffeine is consumed before a workout, there is a significant increase in power output as well as increased capacity for both anaerobic and aerobic exercise.

In short, caffeine makes for a better workout because it combats fatigue and helps you push yourself harder. However, there is a caveat: if you have been consuming caffeine for a long time or in large amounts (through coffee, tea or energy drinks), then your body won't respond as well. This is called 'adaptation effect', which essentially means that all these stimulatory effects can go away when caffeine is used over a period of time.

It's a popular belief that caffeine promotes fat loss. Research shows that caffeine indirectly promotes fat loss, as it blunts hunger and increases energy expenditure.

Dosage, frequency and timing: 1–3 mg/kg a day helps combat tiredness. 4–6 mg/kg taken about an hour before a workout can improve performance in the gym. Find your own tolerance limits and sweet spot. Taking occasional ‘caffeine breaks’ may also help reduce the adaptation effect.

Since caffeine is a stimulant, don’t take it too late in the day or it might interfere with your sleep patterns.

Can you take it every day? Yes, you can.

Is it safe for long term use? Yes, it is.

Possible side-effects: Caffeine is a diuretic which means it causes water loss from the body. As a general rule, stay hydrated. Also, some people who consume pre-workout tend to get a slightly itchy sensation in the skin. Stick to regular black coffee and you’ll be fine.

vi. Vitamin D3

Even though Indians live in a sub-tropical country, most of us are Vitamin D deficient. Surprising, given how much sun we have in India. But as more and more of us spend our days working indoors, we simply aren’t getting enough sunlight which is vital for the synthetization of Vitamin D.

What it is: Vitamin D3 or Cholecalciferol is a fat soluble nutrient and also the most bioavailable form of Vitamin D which makes it the best form of Vitamin D to take as a supplement. However, before doing so, it is important to get yourself tested to check if you actually do have a deficiency. A physician would be the best person to consult if you want to start supplementing D3. Common dietary sources include meat, fatty fish and eggs.

How it works and benefits: Vitamin D deficiency has been linked to osteoarthritis, cancer, hypertension and quite a few autoimmune diseases. Thus, it is worthwhile to ensure that you get sufficient amounts of Vitamin D which a supplement should be able to provide.

Dosage, frequency and timing: You would need to take pretty large doses of Vitamin D3 for a sustained period of time to suffer toxicity. However, as suggested earlier, always consult a physician before supplementing D3.

Can you take it every day? Yes. It’s best taken with a meal or with a source of good fats. Some people also take it weekly. Consult your doctor to know what works best for you.

Is it safe for long term use? Consult a doctor.

Possible side-effects: Very high doses of Vitamin D can cause health issues such as toxicity but it’s rare because you’d have to take enormously large doses for a period of time.

vii. Vitamin C

Remember when we spoke about scurvy? That is caused by a severe Vitamin C deficiency. But there are several other reasons why Vitamin C should be a part of your supplement stack. Almost all detox therapies are useless and a waste of money, except Vitamin C, which is extremely

effective.

Think of what happens to your car when you put in poor quality fuel or petrol that has impurities. It sputters and emits black smoke and, over time, can corrode the fuel tank and reduce your car's useful life. Vitamin C is like the additive that removes impurities and helps the car function smoothly.

What it is: Vitamin C is a water-soluble vitamin and is also known as L-ascorbic acid.

How it works and benefits: Smoking, exposure to pollution, obesity, consuming a diet that's high in sugars and processed foods, etc. cause oxidative stress and produce what are called free radicals. They cause aging and not just of skin and hair. Oxidative stress opens up the body to all kinds of ailments. Vitamin C helps counter or detoxify these free radicals and neutralizes their effects on the body.

Vitamin C's antioxidative and anti-inflammatory properties protect you in a number of ways.

- It is neuroprotective.
- It aids proper blood flow which means that nutrients and oxygen are properly transported throughout the body.
- It regulates cortisol (the stress hormone).
- If you're a man, supplementing with Vitamin C can help preserve your testosterone levels.

You may have also heard that taking a Vitamin C supplement can prevent a cold. While that's not exactly true, it has been found that Vitamin C reduces the duration of a cold by anywhere from 8–14 per cent, provided you take it just when you start developing a cold or if you take it as a daily preventative measure.

Dosage, frequency and timing: Vitamin C is a water-soluble micronutrient, which means that your body does not retain it. You need to replenish stores every single day. The Recommended Daily Intake (RDI) is 100–200 mg. If your diet is on point, then you can easily get this from the food you eat and you may not even need supplements. Higher doses of Vitamin C (up to 2,000 mg) are used by athletes to support their immune systems or for reducing the duration of the common cold.

Can I take it every day? Yes, absolutely.

Is it safe for long term use? Yes, it is.

Possible side-effects: None. (A growing body of evidence indicates that high doses of Vitamin C may have negative effects on a variety of athletic performance tests).

viii. Creatine Monohydrate

If you're serious about lifting weights, building muscle and growing strong, then this is an important discussion. You cannot speak about supplements and not mention creatine, which has to be one of the most polarizing products available for use by athletes and lifters.

That a product with so many extensive research studies to back its efficacy can still be

maligned and misunderstood just shows the absence of a scientific temperament among the fitness community. The simple fact is that creatine deserves to be crowned the king of supplements.

What it is: Creatine is a naturally produced substance within our bodies. It is created from the amino acids glycine, methionine and arginine. Animal products, especially meat and eggs, are a rich source of creatine but the cooking process degrades it to some extent which is why supplementation is a good idea.

Creatine monohydrate is a legal non-hormonal compound that is affordable, effective and can provide your bodies with adequate amounts of creatine which, if you were to rely solely on food sources, would peak at about 1 gram per day. Continual supplementation can bring these levels up to 2–3 grams. Vegetarians, especially, would be greatly benefitted.

How it works and benefits: Creatine improves strength, aids activities that require power and also helps build muscle mass. One of the most effective ways in which creatine works in our bodies is when the concentrations of creatine phosphate increase in our muscles and creatine supplementation greatly assists this.

Research has shown that creatine supplementation helps improve muscle mass, fat free mass, and improves strength in athletes and lifters.

Dosage, frequency and timing: The purpose of creatine supplementation is to saturate the muscle cells with creatine phosphate. There are two ways to do this:

Loading: Consume 20–25 grams of creatine per day for five days, followed by 3–5 grams per day thereafter.

No Loading: Consume 3–5 grams of creatine per day. Usually, with this method, the muscle cells get saturated with creatine in about four weeks.

Loading is usually how beginners are advised to use creatine. However, it is not absolutely necessary and the second method works just fine. Remember, taking creatine won't improve your strength overnight, you will still need to progress in your training to fully reap the benefits.

Timing too isn't so important. Some people report that taking creatine just before their workout helps them feel stronger and lift heavier but that's probably just the placebo effect.

Creatine supplementation has long-term benefits and once you've reached saturation levels, all you need to do is maintain them. It's like filling your fuel tank with petrol and not letting it slip below a certain level.

Can I take it every day? Yes, you can.

Is it safe for long term use? Yes, it is.

Possible side-effects: Make sure you're drinking a little more water than usual if you're supplementing with creatine. This is a simple precaution because creatine retains water. There are no harmful side-effects when you properly use creatine.

2. Ayurvedic/all-natural supplements

Ayurveda has made a comeback in recent years. Looking back, I don't exactly recall what fuelled this renewed interest in the ancient Indian science: was it the Keto crowd and their love for coconut oil, or did it start when coffee chains in the US started selling turmeric soy lattes?

Not wanting to be left behind, Indians have also gone AyurGanic (Ayurveda + Organic), all the while shrugging their shoulders and wagging their 'We told you so!' fingers at the world. Naturally (pun intended), supplement companies have started cashing in on this craze for all things Ayurvedic. They have flooded the market with products that promise to boost your immunity, make you slim for life and turn you into a tiger in the sack. And it's all packaged under the guise of holistic health.

Ayurveda is a powerful system of rejuvenation and has been practised for thousands of years. Used correctly, it can have lasting positive effects on your health. The sad truth is that most supplements that make this claim don't work. Even if they do have some benefits, these are wildly exaggerated. Fortunately, research has been carried out on some of these Ayurvedic ingredients, so that we can decide which ones are worth the money.

Here is a list of Ayurvedic supplements that are supported by science which you may want to include in your daily routine:

i. Ashwagandha

Ashwagandha is one of the principal Ayurvedic herbs and is often called the king of herbs. Men are especially fascinated by it because they've been told that it can boost testosterone levels and give them strength and power that would put even Thor to shame. It's also supposed to give you the virility of a horse in the boudoir which, let's be honest, is probably why there's so much discussion about it!

What it is: Ashwagandha (*Withania somnifera*) is a commonly used medicine in traditional Ayurveda which classifies it as a rejuvenator. Generally, the extract of the root is used for medicinal purposes. It gets its name from the unique horse-like smell that it gives off. Since ancient times, Ashwagandha has been used for increasing energy and gaining relief from stress, among other uses. It is also known to have benefits in alleviating Vata, as a nervine tonic, an adaptogen and as a stress modulator.

These days, it's available as a powder or capsule but if you prefer more traditional preparations, it's also available as a leham (a preparation of ghee, spices and other herbs in addition to Ashwagandha).

But since most of the scientific research has been conducted on modern preparations, we'll stick to that.

How it works and benefits: Modern medicine classifies Ashwagandha as an adaptogen, which means that it relieves stress and significantly reduces its effects, such as fatigue and cognitive impairment. Research shows that it is more effective for people suffering from chronic stress and anxiety disorder. As people who want to get fit and stay healthy, it's important to note that Ashwagandha has been shown to increase testosterone in men who do strength training as well as those men who are infertile. Another important benefit shown by studies is that Ashwagandha

can help reduce overall cholesterol levels. Research also indicates that it can help reduce blood glucose levels.

According to Ayurveda, Ashwagandha has several other benefits such as:

- Improvements in general mood, libido and brain function
- Better skin and hair due to its antioxidant properties
- Cancer-killing properties as it generates reactive oxygen which kill cancer cells
- Protects the heart

Effect on doshas: Ashwagandha helps alleviate Vata and Kapha dosha in the body and is slightly hot in potency.

Dosage, frequency and timing: A low dosage of 50–100 mg is effective for reducing stress. Some people prefer larger doses of 300–500 mg but it's not clear if this has any additional benefits.

Can I take it every day? Going by available research, there does not seem to be any harmful side-effects if you take Ashwagandha daily.

Is it safe for long term use? More research is needed on the long-term effects of Ashwagandha. In Ayurvedic practice, it is considered safe for long-term use.

Possible side-effects: None. However, some people with hot (Pitta) constitution may experience constipation or heat related complaints at a higher dose. In case you have any heat related complaints, it is always better to consult an Ayurvedic physician for the suitable dosage.

ii. Safed Musli

Staying on the topic of testosterone and libido for a bit longer, here's an Ayurvedic supplement that you probably already know but by another name—Safed Musli.

What it is: When Ayurveda was being studied in the forest ashrams of ancient India, an extremely rare herb was discovered, what is now called by the scientific name *Chlorophytum Borivilianum*.

How it works and benefits: This plant has no colour of its own but could make sure your love life was very colourful indeed! It was probably due to this property that the plant was worth its weight in gold and got its name: Safed Musli meaning 'white gold'.

We now know that Safed Musli is almost as potent as Viagra—at least in rats! Hence its other name, herbal Viagra. More research needs to be conducted to see if the same holds true in humans. But preliminary findings are quite promising.

Other potential health benefits include anti-inflammation, stress relief and an aid for diabetes.

Effect on doshas: Musli helps alleviate Vata dosha in the body and is hot in potency.

Dosage, frequency and timing: Since most studies have been conducted in rats, science doesn't tell us a human-appropriate dosage. Ayurveda prescribes Musli to be consumed as 5–6 g of

powder with milk on an empty stomach for its rejuvenating effect or at bedtime for aphrodisiac effects.

Can I take it every day? According to Ayurveda, yes.

Is it safe for long term use? More studies are required but Ayurveda considers it safe.

Possible side-effects: None, as per Ayurveda. Rat studies haven't shown any negative side-effects. However, some people with hot (Pitta) constitution may experience constipation or heat related complaints at a higher dose. In case you have any heat related complaints it is always better to consult an Ayurvedic physician for the suitable dosage.

iii. Turmeric/Curcumin

No list of ayurvedic supplements would be complete without curcumin or what we commonly call 'turmeric'. Along with yoga and the Taj Mahal, turmeric is probably the next thing that comes to the mind when the West thinks of India. But are those overpriced turmeric lattes really worth it?

What it is: Turmeric needs no introduction. Also called Indian saffron, it has been around for over 4,000 years and is a very important spice that's uniquely Indian. No kitchen or religious ceremony is complete without it. Our grandmothers have been making us drink haldi ka doodh, especially during the winter months, as it's considered an excellent tonic for a bad throat and cold.

Turmeric contains several compounds, of which curcumin is the principal bioactive substance that also gives it the beautiful yellow colour. It's a polyphenol and has several properties which is why it has caught the eye of the scientific community.

How it works and benefits: Research has found that curcumin has anti-inflammatory and antioxidant properties. It has also been found to relieve depression, anxiety and pain caused by osteoarthritis. The main issue with curcumin is that it has very poor bio-availability which means that our bodies can't efficiently absorb it. It's been suggested that curcumin taken with piperine (extracted from black pepper) helps improve its absorption.

Turmeric contains a number of other chemicals but more research is needed on their benefits.

Effect on doshas: Turmeric helps alleviate Pitta and Kapha dosha in the body and is slightly hot in potency.

Dosage, frequency and timing: As a condiment or spice, we know how much turmeric to add to our foods. Turmeric and curcumin supplements (mostly capsules) are becoming popular. Studies recommend 1,500 mg of curcumin along with 60 mg of piperine per day (to be taken with food). However, it must be noted that there are some concerns with bioavailability.

Can I take it every day? Yes, you can.

Is it safe for long term use? We've been using turmeric in our cooking for millennia and we're still here!

Possible side-effects: There's no research that suggests that curcumin supplements have any undesirable side effects. Some people may experience nausea or an upset stomach.

3. Supplements that should be avoided

Frankly, this section could be made into a separate book by itself! There are so many BS peddlers and snake-oil salesmen in the fitness industry, it makes you want to give up on the human race altogether! Here, I'll restrict myself to the top offenders.

i. Fat burners

I have lost count of the number of clients who have come to me and asked me if they should take fat burners to lose weight. Invariably, it is a gym trainer or a 'friend' who told them that it'll make them get ripped, or svelte and sexy, or whatever result it is that they want.

The people who ask this question are all, without exception, obese. It used to break my heart to see the look of hope on their faces: they had all probably tried several things to lose weight and nothing had worked. In desperation, they were now at the end of their rope.

Sorry to burst your bubble, people, but a fat burner will burn a hole in your pocket faster than it burns fat. Fat burners are not meant to be taken by obese people who want to lose fat quickly. They can be helpful to people who are already sub-10 per cent body fat and who want to stay that way but even those guys need to eat right and work out regularly.

There are no shortcuts: a caloric deficit and a regular workout schedule are still the only way you can lose fat and achieve your fitness goals! Don't fall for those ads that show a fat guy popping some pills and hey presto, he's now a ripped dude and surrounded by bikini-clad models. It's not going to happen.

If you're reading this book, then just be patient and use the principles that are covered here. Be patient and you'll get there.

ii. Mass gainers

On the other end of the BS spectrum are supplements that promise you the opposite: gain bulk like the Hulk! I must admire the creativity of the people who come up with the names of these products, because it must be really tough trying to convey 'we will stuff you with calories and make you put on weight' through the name of the brand.

That's essentially what mass gainers do; they are simply powders that have a high calorie count, sometimes even as much as 1,000 calories per scoop. All the hard-gainers (those Ectomorphs who find it difficult to gain weight) swoon at the very mention of mass gainers and start visualizing their future selves looking like the Rock or at least a fat, well-to-do zamindaar. There is nothing magical about mass gainers. With just a few tweaks to your diet and workouts, you can easily pack on the pounds.

If you want to gain weight, you know you will need to eat at a caloric surplus, i.e., above your TDEE. Here's how you can do this: first, set your minimum protein macros. Then, add carbohydrates and fat to your diet chart in such a way that your total calories exceed your TDEE.

How many more calories and which macronutrient you want to increase depends on your preference and also on your current state of health. If you like potatoes and rice, by all means, add more carbohydrates to your diet. Pick any high carb diet template. Alternatively, you can add fats.

Usually, I advise people to not drink their calories. But if you want to put on weight, then you can drink things such as Tang, fruit juices and the like, which are loaded with calories but are easy to consume. I know people who consume an entire pack of Glucon D in one day!

But remember, you don't want to put on fat, you want to gain lean mass. This means that along with this hypercaloric diet, you also need to be strong at the gym and lift heavy. That's it! There's your mass gain strategy. And you did it all without shelling out money for an over-priced supplement.

I could really go on and on but I'll stop myself here. I have faith in you guys and know that you will not fall for these marketing gimmicks anymore!

iii. Branched Chain Amino Acids (BCAA)

BCAAs are probably one of the most controversial supplements available in the market, almost as much as creatine. The controversy is not around the ill-effects or otherwise but rather around the question: do we really need BCAA?

What it is: In the chapter on nutrition, I talked about essential and non-essential amino acids. Well, BCAA contain three of those nine essential amino acids: leucine, isoleucine and valine.

There are people who swear by BCAA; their lives would be seriously incomplete if they didn't sip this watermelon/berry/orange flavoured drink every day. They are absolutely certain that they would lose their muscles if they stopped taking BCAA. (I think they just like the taste!)

Research is a little more sceptical about this: yes, BCAA have three essential amino acids. But guess what has all nine essential amino acids? Protein-rich food. As long as you're consuming adequate amounts of quality protein, you're covered!

And if there is a supplement you want to take, there's always good ol' whey protein which, incidentally, also contains all nine essential amino acids!

How it works and benefits: BCAA isn't all bad. There are studies that have shown that they help maintain lean body mass. Other studies suggest that they can help with recovery and help you train harder. For some people, the benefit lies in the fact that BCAA are very low in calories and hence, they won't end up exceeding their daily macros.

Do you need them: You could consider it if you're finding it difficult to complete your protein macros and are worried about muscle loss. However, your first line of defence is always your diet: concentrate on getting good quality, complete protein from foods and your requirement of essential amino acids will be met. If your goal is to build as much muscle as you can and you don't want to take any chances, then go ahead and sip this over-priced supplement.

A Final Word on Supplements

Proper nutrition is still the key to good health and your best bet against deficiencies. Neglecting your diet and depending on supplements alone is like spending all your money and hoping someone gives you a loan. You still might get one but why put yourself in that situation?

If you want to be healthy, fit and live a disease-free life, then nutrition and exercise should be your primary courses of action, not supplements.

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BONUS CHAPTER: CONTINUOUS LEARNING

There's an old saying: if you give a man a fish, you feed him once. But teach him how to fish, and you'll feed him for life. Fitness is something like that. I hope that you, the reader, see this book not as a one-off set of instructions but rather, as a call to action and learning.

I truly believe that people who stop learning stop growing. This book is a great place for you to start your fitness journey and while I've covered a lot of things about fitness and nutrition, it's far from everything.

Don't get me wrong, this book is more than enough to get you started and help you reach your desired fitness goals. But if you keep an open mind and keep learning, you'll be well on your way to becoming an expert, someone who can take complete charge of their health and life. For the curious minds among you, let me suggest a few good places to get started.

Khan Academy

Even though I don't have a formal degree in biology, I know enough to understand and deduce nutritional and physical needs for population groups. This couldn't have been possible without all the free education that I received from Khan Academy. Whether you're learning about the 101 of cellular respiration or epigenetics, Khan Academy has excellent lectures that explain such complex concepts quite simply.

Examine.com

I did cover a few supplements in the book but that was by no means an exhaustive list. As you progress on your fitness journey, there will always be instances when you'll pick up a supplement and wonder, 'Maybe this works.' Instead of reading product reviews (which won't say anything negative) or going by what's claimed on the labels, it'd be a lot wiser to check Examine.com. Want to know if Garcinia Cambogia actually works? Examine. Want to know about testosterone boosters and whether you should really take them? Examine.

The site was started by Kamal Patel and has become *the* go-to place for serious fitness buffs who want to base their supplement choices on science, not hearsay.

Pubmed

Pubmed is the search engine for all things research. If there's new research in any field, it's likely to be listed in the Pubmed directory. Most of the facts quoted in this book can also be cross referenced by going through multiple meta analyses on Pubmed.

A meta analysis is essentially a study of all the existing research studies and is generally considered a good standard for deducing practical conclusions (only behind Cochrane reviews which take a lot of time before they become part of the mainstream recommendations and are primarily taken into consideration for the medical fraternity).

Monthly Applications in Strength Sport (MASS)

For a deeper dive into the latest research and studies, MASS is a great resource. Every month, it covers plenty of brand new nutrition research, and subscribers can actually get continuing education credits for a number of internationally accredited certifications.

Whom should you NOT turn to for advice?

One of the worst things you can do is turn to the media for getting any updates on nutrition and fitness. The media's primary job is not to propagate information but to create more clickbait content as that's how they make money. They love to sensationalize everything and in a lot of cases, you'd be better off taking their advice with a heap of salt. So, if there's a new observational data around how wine improves health or how smoking can help you live for 100 years, you can count on media to report this 'groundbreaking scientific discovery' because there's a good chance that you'll click on the link. Who wouldn't, right?

That's also why most people suffer. They suffer from inherent biases. They're so gullible and easy to trick. Just tell them something they want to hear and they'll put you on a pedestal: Politics 101.

So, there you have it. My two-cents worth on the best resources for learning. Again, this isn't an exhaustive list. As you read deeper and wider about nutrition and exercise science, you'll soon be able to tell the wheat from the chaff.

PARTING WORDS

Thank you for staying with me through the course of this book. It's the end of a beautiful journey...or is it the beginning of a new one?

I said right at the beginning that this book can change your life. I hope that everything you've read so far has convinced you that it is possible to get fit and lead a healthy life.

Remember, it's never too late to make the change. All you need are the right tools and the right mindset.

The book has given you the tools you need. But what about the mindset?

Trust the Process

At Fittr, there's one thing that we always say to our community: trust the process.

Fitness is a journey, not a destination. When you decide that you want to get fit, the path ahead may seem daunting. After all, you've tried so many things in the past to lose fat but nothing seems to have worked. Why should this time be any different?

But don't worry: this time will be different because finally, you'll be doing things right.

There's only one way to stay the course and that is to love the journey and trust the process that has been laid out in this book.

But will it all be smooth sailing? Not always. You will have doubts and not-so-great days. There will be times when you just don't want to exercise or when your day seems so hectic that preparing your meals is the last thing on your mind.

That's perfectly normal. Everyone has days that feel off. The important thing is to not let one bad day turn into a week of poor choices. If you allow yourself to slip, it will take that much more effort to get back on track.

So, what do you do when you simply don't feel like working out or eating right?

Take a day off. And then, come back stronger the next day.

Rome wasn't built in a day, but it was built every single day, brick by brick. That's the attitude you need to bring to your fitness journey.

And if you still need motivation, just open any chapter in this book and read it again. Better yet, read one of the transformation stories. It will remind you again of why you started in the first place and show you a glimpse of what you too can achieve.

If you want to reach out to me, I'm always available for you. Just send an email to jc@fittr.com and let me know how I can help.

As I said, fitness is a journey—and I would love to hear how your journey is shaping up.

You've got this! All the best.